

# Measuring Normative Information

What a value-alignment dataset can and cannot tell you, derived from zero

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2026-08-01

## Abstract

A public dataset of human value judgments ships 15,248 written criteria with numerical weights, and 82.3% of those criteria contain no condition, no exception, no priority, no hedge and no prohibition – the collected normative payload is content, a sign, and a magnitude. This paper asks what can be measured about such an object at all, and answers it as a typed question: every quantity is defined relative to a declared query family, a declared decision family, and a named instrument, and each is derived here from nothing, assuming no statistics. What survives is: (i) a theorem that no finite weight encodes a veto, because the penalty reproducing one depends on which candidate responses happened to be shown, with the counterexample present in the data at  $2.65\times$  the elicitation scale; (ii) that a person’s own written criteria predict a *stranger’s* ranking at 0.310 and their own at 0.393, so the individual signal in the elicitation is 0.083 of cosine and nothing downstream can destroy what was never collected; (iii) that comparative measurements are invariant across three executors and a presentation-order swap (Spearman 0.963–1.000) while absolute ones are not (40% of winners change with the reader, 8 points with the page order); and (iv) that one bit per criterion – the sign alone – reproduces 89.8% of the shipped decisions. Six claims were withdrawn during this work and are listed with what killed them; five of the six died the same way, to a verdict decided by a constant the author chose rather than by a comparison against a range of defensible references. The central claim ends if a person’s induced ordering proves unstable across response sets: what was elicited would then be a property of the probe rather than of the person, and no measurement downstream would be about values at all.

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# 1 Introduction

## 1.1 The question, and why the obvious version of it is the wrong one

The obvious question is: *how much of what people said survives the pipeline?* It has no answer, and the reason is not empirical.

To say that two normative objects carry “the same information” one must fix what one intends to ask of them. [Proposition 131](#) shows that the choice of question family determines the answer completely: under one admissible family every change is total loss, under another no change is any loss. A framework that reports a single percentage has therefore not measured a property of the pipeline; it has reported a choice it did not disclose.

The question this paper answers instead is: *given a declared family of decisions, what is identified, what is bounded, and what is forbidden?* That version has answers, several of them are theorems rather than measurements, and the largest of them turns out to concern the elicitation rather than the pipeline.

## 1.2 Scope, identity, and contributions

The object is a released dataset of human comparative judgments over four candidate responses per prompt, together with written criteria and numeric weights, and a compiled “core” standard derived from them (§5). The satisfaction of each criterion by each response is not in the release; it is reconstructed with a language-model judge, and every consequence of that is tracked explicitly (§10).

Contributions, in the order they appear:

1. A self-contained derivation of every statistical tool used, from the definition of probability through the design effect, attenuation and minimum detectable effect (§3). No external statistical background is assumed at any point.
2. The probe bound ([theorem 94](#)): the elicitation returns two free parameters per person per prompt, whatever the criteria say.
3. A measurement that the weight scale is used as an *interval* scale, with the fraction of decisions that depend on it (§6.2).
4. The social-choice structure of the corpus, including the finding that the weighted-criteria route agrees with the Condorcet winner only 68.4% of the time while every ranking rule agrees 97–98% (§7).
5. [theorem 110](#) and [theorem 111](#): the exact condition under which a penalty reproduces a veto, and the proof that no finite penalty does so uniformly.
6. Normalised regret ([definition 133](#)) as the decision-theoretically correct estimand, with the exact identity showing the flip rate to be its coarsening ([lemma 135](#)).
7. The gauge result separating comparative from absolute claims (§10.3).
8. An explicit non-identification section with proofs (§17), and a design specification derived line by line from it (§23).

### 1.3 Relation to neighbouring work

Everything in §3 is standard and is reproduced only so the paper is self-contained. Blackwell’s comparison of experiments, Arrow’s impossibility theorem, Sen’s liberal paradox, the Shapley value, the Spearman–Brown prophecy formula, the design effect and classical attenuation are all classical; they are derived rather than cited because a reader with no statistical background cannot otherwise check the argument. The novelty boundary is stated explicitly in §19 and the Discussion: what is new is the application to this object and five specific measured quantities, not the machinery.

### 1.4 Roadmap

§3 builds the mathematics. §5 defines the object exactly. §13.1 states the measurement contract. §6–§10 walk the chain one arrow at a time. §15 and §16 give the estimators. §17 proves what cannot be identified. §18 states the findings with their confidence cards, §19 what did not survive, §20 what would end each claim, and §23 what a sufficient design requires.

## 2 Every symbol, from arithmetic upward

This section defines everything the paper uses. It assumes arithmetic — adding, multiplying, and the idea of a number — and nothing else. A reader who is fluent may skip to §3; a reader who is not should be able to finish this section and then check every subsequent line.

### 2.1 Numbers, inequalities, intervals

$\mathbb{R}$  denotes the real numbers: everything on the number line, including fractions and irrationals like  $\sqrt{2}$ .  $a < b$  means  $a$  lies to the left of  $b$ .  $[a, b]$  is the set of all  $x$  with  $a \leq x \leq b$  (endpoints included);  $(a, b)$  excludes the endpoints.  $|x|$  is the *absolute value*:  $x$  if  $x \geq 0$ , and  $-x$  otherwise. It is the distance from  $x$  to 0, so it is never negative.

*One fact used repeatedly.*

Multiplying an inequality by a *positive* number preserves it; multiplying by a *negative* number **reverses** it.  $2 < 3$  but  $-2 > -3$ . §3.10 depends on this at one specific step, and the step is marked there.

### 2.2 Functions

A *function*  $f$  from a set  $\mathcal{A}$  to a set  $\mathcal{B}$ , written  $f : \mathcal{A} \rightarrow \mathcal{B}$ , assigns to each element of  $\mathcal{A}$  exactly one element of  $\mathcal{B}$ .  $\mathcal{A}$  is the *domain*,  $\mathcal{B}$  the *codomain*. “Exactly one” is the whole content: a rule that sometimes gives two answers is not a function.

**Definition 1** (Monotone).  $f$  is *strictly increasing* if  $x < y$  implies  $f(x) < f(y)$ . Such a function preserves order and nothing else: it can stretch and squeeze distances freely. §6.2 uses exactly this property to test whether the paper’s central data are ordinal or not.

### 2.3 Sum and product notation

$$\sum_{i=1}^n x_i = x_1 + x_2 + \cdots + x_n, \quad \prod_{i=1}^n x_i = x_1 \cdot x_2 \cdots x_n.$$

The letter  $i$  is a *dummy*:  $\sum_{i=1}^3 x_i$  and  $\sum_{j=1}^3 x_j$  are the same number. Three manipulation rules are used later and each is just ordinary arithmetic written compactly.

*Rule 1 — pulling out a constant.*

$$\sum_i (c x_i) = c \sum_i x_i, \text{ because } cx_1 + cx_2 + \cdots = c(x_1 + x_2 + \cdots).$$

*Rule 2 — splitting a sum.*

$$\sum_i (x_i + y_i) = \sum_i x_i + \sum_i y_i, \text{ because addition can be reordered.}$$

*Rule 3 — reordering, and why it is safe here.*

For a *finite* collection, adding the terms in any order gives the same total. Every sum in this paper is finite, so grouping terms differently — for instance grouping outcomes by the value a function takes on them, as in the proof of lemma 34 — never changes the answer. (For infinite sums this is false in general; the paper never needs one.)

**Definition 2** (Factorial).  $n! := 1 \cdot 2 \cdot 3 \cdots n$ , with  $0! := 1$  by convention. It counts the number of ways to arrange  $n$  distinct objects in a row:  $n$  choices for the first, then  $n - 1$ , and so on.  $3! = 6$ , and the six orderings are listed explicitly in §11.

## 2.4 Powers, roots, logarithms

$x^2 := x \cdot x$ ;  $\sqrt{x}$  is the non-negative number whose square is  $x$ ;  $x^{1/2} = \sqrt{x}$ . The *logarithm*  $\log x$  is the power to which the base must be raised to give  $x$ . Only two properties are used:  $\log(ab) = \log a + \log b$  and  $\log$  is strictly increasing. The second is why correlating log-transformed quantities in [remark 93](#) is a statement about *proportional* rather than absolute co-movement.

## 2.5 Max, min, argmax, sup, inf

$\max_i x_i$  is the largest of the  $x_i$ ;  $\min_i x_i$  the smallest.

**Definition 3** (argmax).  $\arg \max_i x_i$  is the *index* at which the maximum occurs, not the value. If  $x = (4, 9, 2)$  then  $\max = 9$  and  $\arg \max = 2$ . When two entries tie, the  $\arg \max$  is not unique; the paper's decision rules break ties by index order, and exact ties in continuous scores occur with probability zero in the data.

**Definition 4** (Supremum).  $\sup$  is the least upper bound: the smallest number that is  $\geq$  every element of a set. It coincides with max when the maximum is attained, and exists even when it is not.  $\sup\{1 - \frac{1}{n} : n \geq 1\} = 1$  although 1 is never attained. [theorem 111](#) states  $\sup_X \Delta(X) = \infty$ , meaning: for any bound you name, some candidate set exceeds it.

## 2.6 Sets, and the four operations used

$\mathcal{A} \cup \mathcal{B}$  (union) contains everything in either;  $\mathcal{A} \cap \mathcal{B}$  (intersection) everything in both;  $\mathcal{A} \subseteq \mathcal{B}$  (subset) means every element of  $\mathcal{A}$  is in  $\mathcal{B}$ ;  $\emptyset$  is the empty set;  $|\mathcal{A}|$  is the count of elements;  $\mathcal{A} \setminus \mathcal{B}$  removes from  $\mathcal{A}$  everything that is in  $\mathcal{B}$ .

**Lemma 5** (Union bound, the pointwise version).  $\mathbf{1}\{x \in \bigcup_i \mathcal{A}_i\} \leq \sum_i \mathbf{1}\{x \in \mathcal{A}_i\}$ .

*Proof.* If  $x$  is in none of the sets, both sides are 0. If  $x$  is in at least one, the left side is 1 and the right side is at least 1 because it counts the sets containing  $x$ . There is no third case.  $\square$

[Theorem 78](#) is this lemma plus taking expectations, and nothing else.

## 2.7 Tuples and vectors

An *ordered tuple*  $(x_1, \dots, x_k)$  is a list in which position matters:  $(1, 2) \neq (2, 1)$ . The set of all such lists of  $k$  real numbers is written  $\mathbb{R}^k$ , and an element of it is a *vector*.

Throughout this paper  $k = 4$  and the four positions are the four candidate responses  $A, B, C, D$ . So a vector such as  $(5.1, 7.1, 5.7, 2.8)$  means: this norm scores response  $A$  at 5.1,  $B$  at 7.1, and so on.

**Definition 6** (Addition and scaling).  $(x_1, \dots, x_k) + (y_1, \dots, y_k) := (x_1 + y_1, \dots, x_k + y_k)$  and  $c(x_1, \dots, x_k) := (cx_1, \dots, cx_k)$ . Both act position by position.

*Why this is the right operation here.*

A criterion with weight  $w$  and satisfaction profile  $s = (s_A, s_B, s_C, s_D)$  contributes  $w s$  to the score, and several criteria contribute their sum. That is exactly [definition 91](#), and it is the only place vector arithmetic enters the construction.

## 2.8 The dot product, the norm, and the angle

**Definition 7** (Dot product).  $x \cdot y := \sum_{i=1}^k x_i y_i$ , a single number.

**Definition 8** (Norm).  $\|x\| := \sqrt{x \cdot x} = \sqrt{\sum_i x_i^2}$ .

*Why the norm is the length.*

In two dimensions,  $x = (x_1, x_2)$  is the corner of a right triangle with legs  $x_1$  and  $x_2$ ; Pythagoras gives hypotenuse  $\sqrt{x_1^2 + x_2^2}$ , which is  $\|x\|$ . In  $k$  dimensions the same formula is taken as the *definition* of length, and it behaves as length should:  $\|cx\| = |c| \|x\|$ , and  $\|x\| = 0$  only when every coordinate is 0.

**Theorem 9** (The dot product measures angle). *For non-zero  $x, y$ ,*

$$\cos \theta = \frac{x \cdot y}{\|x\| \|y\|},$$

*where  $\theta$  is the angle between  $x$  and  $y$ .*

*Proof. Step 1.* The law of cosines for the triangle with sides  $\|x\|$ ,  $\|y\|$  and  $\|x - y\|$  states  $\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\|\cos\theta$ .

*Step 2.* Expand the left side using the definition:

$$\|x - y\|^2 = \sum_i (x_i - y_i)^2 = \sum_i x_i^2 - 2 \sum_i x_i y_i + \sum_i y_i^2 = \|x\|^2 - 2(x \cdot y) + \|y\|^2.$$

*Step 3.* Set the two expressions equal. The terms  $\|x\|^2$  and  $\|y\|^2$  cancel from both sides, leaving  $-2(x \cdot y) = -2\|x\|\|y\|\cos\theta$ .

*Step 4.* Divide by  $-2\|x\|\|y\|$ . □

**Definition 10** (Cosine similarity).  $\cos(x, y) := \frac{x \cdot y}{\|x\|\|y\|}$ , a number in  $[-1, 1]$ . +1 means same direction, 0 perpendicular, -1 opposite.

*Remark 11* (The single most important consequence for this paper). Cosine similarity is *blind to length*:  $\cos(x, y) = \cos(2x, y)$  for any scaling. That is precisely why it is the right tool once [remark 93](#) establishes that the length of  $N_i$  measures how many criteria a person happened to rate rather than anything normative.

*A worked cosine.*

$x = (1, 0)$ ,  $y = (1, 1)$ . Then  $x \cdot y = 1$ ,  $\|x\| = 1$ ,  $\|y\| = \sqrt{2} = 1.4142$ , so  $\cos = 1/1.4142 = 0.7071$ , which is  $\cos 45^\circ$ . Correct:  $(1, 1)$  points at 45 degrees to  $(1, 0)$ .

## 2.9 Centring, and what it removes

**Definition 12** (Centred vector).  $\tilde{x} := x - \bar{x} \mathbf{1}$  where  $\bar{x} = \frac{1}{k} \sum_i x_i$  and  $\mathbf{1} = (1, 1, \dots, 1)$ . Equivalently, subtract the average from every entry.

*Two facts, both immediate.*

$\sum_i \tilde{x}_i = \sum_i x_i - k\bar{x} = 0$ ; and if  $y = x + c\mathbf{1}$  then  $\tilde{y} = \tilde{x}$ , because the shift moves the average by exactly  $c$  as well. Centring erases the common level and keeps only the contrasts, which is exactly what the decision uses.

## 2.10 Matrices, rank, and the one matrix this paper needs

A *matrix* is a rectangular array of numbers;  $M_{ij}$  is the entry in row  $i$ , column  $j$ . A matrix acts on a vector by  $(Mx)_i := \sum_j M_{ij}x_j$ : each output coordinate is a weighted sum of the input coordinates.

**Definition 13** (Linear independence, span, rank). Vectors  $v_1, \dots, v_m$  are *linearly independent* if the only way to make  $\sum_j c_j v_j = 0$  is to take every  $c_j = 0$  — none of them is redundant. Their *span* is the set of all  $\sum_j c_j v_j$ . The *rank* of a matrix is the number of linearly independent vectors among its outputs: the dimension of the space it can reach.

**Lemma 14** (The centring matrix and its rank). Let  $C := I - \frac{1}{k} \mathbf{1}\mathbf{1}^\top$ , so that  $Cx = \tilde{x}$ . Then  $C$  is a *projection* ( $C^2 = C$ ) and  $\text{rank}(C) = k - 1$ .

*Proof. Step 1 — C does centre.*  $(\mathbf{1}^\top x)_i = \sum_j x_j$  for every  $i$ , so  $\frac{1}{k} \mathbf{1}\mathbf{1}^\top x = \bar{x} \mathbf{1}$ , and  $Cx = x - \bar{x} \mathbf{1}$ .

*Step 2 — it is idempotent.* Centring an already-centred vector changes nothing, since its average is 0. So  $C(Cx) = Cx$ , i.e.  $C^2 = C$ .

*Step 3 — what it kills.*  $C\mathbf{1} = \mathbf{1} - \mathbf{1} = 0$ , so the one-dimensional space of constant vectors is sent to zero.

*Step 4 — what it keeps.* If  $Cx = 0$  then  $x = \bar{x} \mathbf{1}$ , i.e.  $x$  is constant. So the set sent to zero is *exactly* the constants, of dimension 1.

*Step 5 — count.* The input space has dimension  $k$ ; a one-dimensional subspace is destroyed; the image therefore has dimension  $k - 1$ . For  $k = 4$ : rank 3. □

[Lemma 14](#) is the entire content of the first half of [theorem 94](#).

## 2.11 Equivalence, quotients, and the sphere

**Definition 15** (Equivalence relation). A relation  $\sim$  that is reflexive ( $x \sim x$ ), symmetric ( $x \sim y \Rightarrow y \sim x$ ) and transitive ( $x \sim y, y \sim z \Rightarrow x \sim z$ ). It partitions a set into disjoint *equivalence classes*:  $[x] := \{y : y \sim x\}$ .

The two used here.

(i)  $u \sim u + c\mathbf{1}$ : utilities differing by a constant, which decide identically. (ii)  $u \sim \lambda u$  for  $\lambda > 0$ : utilities differing by positive scale, which order identically.

**Definition 16** (Quotient). The set of equivalence classes, written  $\mathcal{A}/\sim$ . Its dimension is the dimension of  $\mathcal{A}$  minus the dimension of what was collapsed.

**Definition 17** (Sphere).  $S^m := \{x \in \mathbb{R}^{m+1} : \|x\| = 1\}$ , the set of unit-length vectors in  $\mathbb{R}^{m+1}$ . It has dimension  $m$ : one constraint removes one degree of freedom.  $S^1$  is a circle in the plane;  $S^2$  is the surface of a ball in three dimensions.

Putting the two together, which is [theorem 94](#).

Starting from  $\mathbb{R}^4$ : quotient by constants leaves dimension 3; the unit-length representatives of that 3-dimensional space form  $S^2$ , of dimension 2. Hence “two free parameters”.

## 2.12 Order statistics: median, quantile, percentile

Sort the values as  $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$ .

**Definition 18** (Median). The middle value:  $x_{((n+1)/2)}$  if  $n$  is odd, and the average of  $x_{(n/2)}$  and  $x_{(n/2+1)}$  if  $n$  is even. For  $(2, 4, 4, 4, 6)$  it is 4; for  $(2, 4, 4, 6)$  it is  $(4 + 4)/2 = 4$ .

**Definition 19** (Quantile). The  $q$ -quantile is the value below which a fraction  $q$  of the data lies. The median is the 0.5-quantile;  $p_{25}$  and  $p_{75}$  are the 0.25- and 0.75-quantiles (*quartiles*). “ $p_{95} = 0.800$ ” in §15 means 95% of prompts have a relative margin below 0.800.

*Remark 20* (Why the paper reports quantiles beside means). A mean can be moved arbitrarily by one extreme value; a median cannot. Reporting both, as [section 8.2](#) does with median 3.64 and maximum 26.54, exposes a skew that a mean alone would hide — and in that table the *maximum* is the load-bearing number, because [theorem 111](#) is about the worst case.

## 2.13 Rank correlation (Spearman)

**Definition 21** (Spearman  $\rho$ ). Replace each value by its rank (smallest = 1, next = 2, ...) in its own list, then compute the ordinary correlation of the ranks.

Why the paper uses it for the gauge test.

It responds only to *order*, so it answers “do the two instruments rank the operators the same way” without being distracted by their different scales — which is the exact question §10.3 asks. A Spearman of +1.000 means identical ordering; 0 means unrelated ordering.

Worked.

Instrument 1 gives (0.116, 0.051, 0.042, 0.023), ranks (4, 3, 2, 1). Instrument 2 gives (0.139, 0.061, 0.035, 0.025), ranks (4, 3, 2, 1). Identical ranks, so  $\rho = +1.000$ , even though no value matches.

## 2.14 Bits, and what quantisation means

**Definition 22** (Bit). One *bit* distinguishes two possibilities.  $b$  bits distinguish  $2^b$ : one bit gives 2 levels, two bits 4, three bits 8, eight bits 256.

**Definition 23** (Quantisation to  $b$  bits). Given values in  $[\ell, h]$ , replace each  $x$  by the nearest of the  $2^b$  evenly spaced levels  $\ell + \frac{j}{2^b-1}(h - \ell)$ ,  $j = 0, \dots, 2^b - 1$ .

Worked, at one bit.

With weights ranging over  $[-2, 8]$ , one bit leaves the two levels  $\{-2, 8\}$ : every negative weight becomes  $-2$  and every positive one 8. Only the *sign* survives. §9 reports that this reproduces 89.8% of the shipped decisions.

## 2.15 Min–max normalisation

**Definition 24.**  $\hat{x}_i := \frac{x_i - \min_j x_j}{\max_j x_j - \min_j x_j} \in [0, 1]$ , with  $\hat{x} = 0$  at the worst entry and 1 at the best.



Why §7 uses it, and what it assumes.

Comparing utility across people requires putting them on a common scale, and nothing in the elicitation supplies one. Min–max makes every person’s best option worth 1 and worst worth 0 *by construction*, which is a stipulation, not a measurement. It is stated as such: the Blackwell deficiency of 0.1298 is a number relative to that normalisation, and a different interpersonal scale would give a different figure.

## 2.16 Conditional probability and conditional expectation

**Definition 25** (Conditional probability).  $\mathbb{P}(E \mid F) := \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}$  when  $\mathbb{P}(F) > 0$ : restrict attention to the outcomes in  $F$  and renormalise so they sum to 1.

**Definition 26** (Conditional expectation).  $\mathbb{E}[X \mid F] := \frac{\sum_{\omega \in F} X(\omega)\mathbb{P}(\omega)}{\mathbb{P}(F)}$ , the average of  $X$  over  $F$  alone.

**Lemma 27** (Law of total expectation, the two-event case). *For an event  $F$  with complement  $F^c$ ,*

$$\mathbb{E}[X] = \mathbb{E}[X \mid F] \mathbb{P}(F) + \mathbb{E}[X \mid F^c] \mathbb{P}(F^c).$$

*Proof.* Split the defining sum by whether the outcome lies in  $F$ :

$$\mathbb{E}[X] = \sum_{\omega \in F} X(\omega)\mathbb{P}(\omega) + \sum_{\omega \in F^c} X(\omega)\mathbb{P}(\omega).$$

Multiply and divide the first sum by  $\mathbb{P}(F)$  and the second by  $\mathbb{P}(F^c)$ , then apply the definition above to each. (If  $\mathbb{P}(F) = 0$  the term is absent and the identity is trivial.)  $\square$

**Lemma 27** with  $F = \{\text{the decision flipped}\}$  and  $X = \text{Reg}$  is exactly [lemma 135](#), since  $\text{Reg} = 0$  on  $F^c$  so the second term vanishes.

## 2.17 Independent and identically distributed

*Identically distributed* means every observation is drawn from the same  $\mathbb{P}$ . *Independent* means no observation carries information about another. Written *iid*. §3.13 exists because in this data the observations are identically distributed but **not** independent, and only the second half fails.

## 2.18 Markov kernels

**Definition 28** (Markov kernel). A rule  $K$  that, given an input  $z$ , produces a *probability distribution* over outputs, written  $K(\cdot \mid z)$ . A deterministic function is the special case that puts all the probability on one output, written  $\delta_{f(z)}$ .

Why the paper needs it.

**Theorem 119** says a garbling cannot help. Aggregation is deterministic, hence a garbling with the degenerate kernel  $\delta_{A(z)}$ ; the theorem then applies verbatim without any probabilistic aggregation being involved.

## 2.19 Integrals, only as much as $\Phi$ requires

The paper uses the normal distribution function  $\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$ . The integral sign means *area under the curve to the left of  $z$* . The curve  $\frac{1}{\sqrt{2\pi}} e^{-t^2/2}$  is the familiar bell; its total area is 1, so  $\Phi(z)$  is the fraction of the bell lying left of  $z$ .

**Definition 29** (Quantile of the normal).  $z_q$  is the point with  $\Phi(z_q) = q$ . The three used are  $z_{0.975} = 1.9600$ ,  $z_{0.80} = 0.8416$ ,  $z_{0.99} = 2.3263$ . By the symmetry of the bell about 0,  $z_{1-q} = -z_q$ ; this symmetry is the step used in [theorem 65](#).

**Remark 30** (No calculus is needed beyond this). The paper never differentiates or integrates anything.  $\Phi$  enters only as a table of three numbers, and every use of it is an instance of “how far out in the bell is this”.

## 2.20 Sigmoid and logit, for the instrument

The judge produces a number  $\lambda$  (a difference of log-probabilities) and it is mapped to  $(0, 1)$  by the *sigmoid*  $\sigma(\lambda) = 1/(1 + e^{-\lambda})$ , which is strictly increasing with  $\sigma(0) = 0.5$ .

**Boundary.**  $\sigma(\lambda)$  lies in  $(0, 1)$  and is therefore easily mistaken for a probability. It is not one. It is a monotone re-expression of a model’s internal score, calibrated against nothing. Every use of  $S$  in this paper carries that caveat, and §10 measures its consequence rather than assuming it away.

## 2.21 A table of every symbol

| symbol                                                        | meaning                                              | first defined                |
|---------------------------------------------------------------|------------------------------------------------------|------------------------------|
| $\mathbf{1}\{P\}$                                             | indicator, 1 if $P$ true                             | §3                           |
| $\sum, \prod, n!$                                             | sum, product, factorial                              | §2                           |
| $\arg \max$                                                   | the index of the maximum                             | §2                           |
| $\sup, \inf$                                                  | least upper / greatest lower bound                   | §2                           |
| $x \cdot y, \ x\ $                                            | dot product, length                                  | §2                           |
| $\cos(x, y)$                                                  | cosine similarity                                    | §2                           |
| $\tilde{x}$                                                   | centred vector                                       | §2                           |
| $C$                                                           | centring matrix, rank $k - 1$                        | lemma 14                     |
| $S^m$                                                         | unit sphere, dimension $m$                           | §2                           |
| $\mathbb{P}, \mathbb{E}, \text{Var}, \text{Cov}, \text{Corr}$ | probability, mean, variance, covariance, correlation | §3                           |
| $se$                                                          | standard error = $sd/\sqrt{n}$                       | §3.7                         |
| $s^2$                                                         | sample variance, divisor $n - 1$                     | lemma 53                     |
| $\Phi, z_q$                                                   | normal cdf and its quantiles                         | §2                           |
| $\rho$                                                        | intraclass correlation <i>or</i> Spearman            | §3.13, §2                    |
| DEFF                                                          | design effect $1 + (R - 1)\rho$                      | theorem 70                   |
| MDE                                                           | minimum detectable effect                            | theorem 65                   |
| rel                                                           | reliability                                          | §3.17                        |
| $\mathcal{X}, X, k$                                           | response space, probe set, its size (= 4)            | §5                           |
| $S, W, V, \Pi$                                                | satisfaction, weight, veto set, ranking              | definition 89                |
| $Y_P, N_i$                                                    | collective and individual induced scores             | definition 91, definition 92 |
| $A, G$                                                        | aggregation rule and its output                      | equation (7)                 |
| Reg                                                           | normalised regret                                    | definition 133               |
| $\pi$                                                         | flip rate                                            | §15                          |
| $m$                                                           | decision margin                                      | proposition 138              |
| $\Delta$                                                      | the penalty a veto requires                          | theorem 110                  |
| $\phi_i$                                                      | Shapley influence                                    | §16                          |
| $v(S)$                                                        | characteristic function of the coalition game        | §16                          |
| $K(\cdot   z)$                                                | Markov kernel                                        | §2                           |
| $\sigma(\lambda)$                                             | sigmoid                                              | §2                           |

**Table 1:** Every symbol used in the paper.  $\rho$  is overloaded by convention; the two uses never appear in the same expression and are disambiguated at each site.

## 3 The mathematics, from zero

This section assumes no statistics. It builds, in order, every object the rest of the paper uses. Nothing here is specific to the dataset; §5 onward uses only what is defined below. A reader who already knows this material should still read §3.13, §3.12 and §3.17, because those three are where papers in this area go wrong.

### 3.1 The one identity everything is built on

$\mathbf{1}\{P\}$  is the *indicator*: 1 if  $P$  is true, 0 if false. It converts a yes/no fact into a number, and every “what fraction of cases” quantity in this paper is built from it, via the identity that the average of an indicator *is* a proportion:

$$\frac{1}{n} \sum_{i=1}^n \mathbf{1}\{P_i\} = \frac{\text{number of } i \text{ with } P_i \text{ true}}{n}. \quad (1)$$

Every rate reported in this paper – how often a decision changes, how often a veto binds – is an instance of [equation \(1\)](#). That is why they can all be treated with one set of tools.

### 3.2 Finite probability

**Definition 31** (Probability space). A *finite probability space* is a finite set  $\Omega$  of *outcomes* together with a function  $\mathbb{P}$  assigning to each  $\omega \in \Omega$  a number  $\mathbb{P}(\omega) \geq 0$ , such that  $\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1$ . For a subset  $E \subseteq \Omega$  (an *event*),  $\mathbb{P}(E) := \sum_{\omega \in E} \mathbb{P}(\omega)$ .

Two consequences, both immediate from the definition and both used later.

*Sub-step 0.2.1 — probabilities are bounded.*

$E \subseteq \Omega$  implies  $\mathbb{P}(E) \leq \sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1$ , and  $\mathbb{P}(E) \geq 0$  because it is a sum of non-negative terms. So  $0 \leq \mathbb{P}(E) \leq 1$  always.

*Sub-step 0.2.2 — disjoint events add.*

If  $E \cap F = \emptyset$  then no outcome is counted twice, so  $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F)$ .

**Definition 32** (Random variable). A *random variable* is a function  $X : \Omega \rightarrow \mathbb{R}$ . It is not random and it is not a variable; it is a rule that reads off a number from whatever outcome occurred. We write  $\mathbb{P}(X = x)$  for  $\mathbb{P}(\{\omega : X(\omega) = x\})$ .

### 3.3 Expectation: the derivation, not the formula

Suppose we observe  $X$  many times and average the results. If outcome  $x$  occurs a fraction  $\mathbb{P}(X = x)$  of the time, then in  $n$  observations it contributes  $x \cdot n\mathbb{P}(X = x)$  to the total. Summing over the possible values and dividing by  $n$ :

$$\frac{1}{n} \sum_{\text{values } x} x \cdot n\mathbb{P}(X = x) = \sum_x x \mathbb{P}(X = x). \quad (2)$$

The  $n$  cancels. The result does not depend on how many times we look. That limit-of-the-average is the definition.

**Definition 33** (Expectation).  $\mathbb{E}[X] := \sum_x x \mathbb{P}(X = x)$ .

**Lemma 34** (Linearity). For random variables  $X, Y$  on the same space and constants  $a, b$ :  $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ .

*Proof.* Write the expectation as a sum over outcomes rather than values, which is equivalent because grouping outcomes by their value only reorders a finite sum:  $\mathbb{E}[X] = \sum_{\omega} X(\omega)\mathbb{P}(\omega)$ . Then

$$\mathbb{E}[aX + bY] = \sum_{\omega} (aX(\omega) + bY(\omega))\mathbb{P}(\omega) = a \sum_{\omega} X(\omega)\mathbb{P}(\omega) + b \sum_{\omega} Y(\omega)\mathbb{P}(\omega),$$

which is  $a\mathbb{E}[X] + b\mathbb{E}[Y]$ . *No independence was used.* This is why linearity is the one tool that never needs an assumption, and it is used at every later step where an assumption would be contentious.  $\square$

**Corollary 35** (Expectation of an indicator).  $\mathbb{E}[\mathbf{1}\{P\}] = 1 \cdot \mathbb{P}(P) + 0 \cdot (1 - \mathbb{P}(P)) = \mathbb{P}(P)$ .

**Corollary 35** is the bridge between “a proportion I measured” and “a probability I want to talk about”, and every rate in this paper crosses it.

### 3.4 Variance: why squared deviation, and not something else

We want a single number for “how spread out is  $X$ ”. Let  $\mu := \mathbb{E}[X]$ .

*Sub-step 0.4.1 — the naive attempt fails.*

$\mathbb{E}[X - \mu] = \mathbb{E}[X] - \mu = 0$  by [lemma 34](#), always. Average deviation is identically zero and carries no information: positive and negative deviations cancel exactly.

*Sub-step 0.4.2 — two repairs are available.*

Remove the sign either by absolute value,  $\mathbb{E}[X - \mu]$ , or by squaring,  $\mathbb{E}[(X - \mu)^2]$ . Both are legitimate measures of spread. Squaring is chosen for one reason that matters here and one that does not.

- **Matters:** squared deviations *add* for independent quantities ([lemma 42](#)), so the spread of an average has a closed form. Absolute deviations do not add.
- **Does not matter:** differentiability, which is a convenience for optimisation and irrelevant to this paper.

**Definition 36** (Variance and standard deviation).  $\text{Var}(X) := \mathbb{E}[(X - \mu)^2]$  and  $\text{sd}(X) := \sqrt{\text{Var}(X)}$ . The square root exists because  $\text{Var}(X)$  is an average of squares, hence  $\geq 0$ . The standard deviation is in the same units as  $X$ ; the variance is in squared units, which is why intervals are reported with the former.

**Lemma 37** (Computational form).  $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ .

*Proof.*  $\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mu^2$ , using [lemma 34](#) and the fact that  $\mu$  is a constant.  $\square$

**Corollary 38** (Variance of an indicator). *If  $X = \mathbf{1}\{P\}$  with  $\mathbb{P}(P) = \pi$ , then  $X^2 = X$  (since  $0^2 = 0, 1^2 = 1$ ), so by lemma 37  $\text{Var}(X) = \pi - \pi^2 = \pi(1 - \pi)$ .*

Corollary 38 is why every yes/no rate in this paper has a variance we can write down without estimating anything extra, and why rates near 0 or 1 are measured more precisely than rates near  $\frac{1}{2}$ .

**Lemma 39** (Scaling).  $\text{Var}(aX + b) = a^2 \text{Var}(X)$ .

*Proof.*  $\mathbb{E}[aX + b] = a\mu + b$ , so the deviation is  $aX + b - (a\mu + b) = a(X - \mu)$ , whose square is  $a^2(X - \mu)^2$ . Take expectations and apply lemma 34. The additive constant  $b$  vanishes because shifting everything does not change spread.  $\square$

### 3.5 Independence, covariance, correlation

**Definition 40** (Independence).  $X$  and  $Y$  are *independent* if  $\mathbb{P}(X = x \text{ and } Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$  for all  $x, y$ : knowing one tells you nothing about the other.

**Definition 41** (Covariance).  $\text{Cov}(X, Y) := \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$ .

The sign of the product  $(X - \mu_X)(Y - \mu_Y)$  is positive when both deviate the same way and negative when they deviate oppositely, so the average is positive for co-movement and negative for counter-movement. Note  $\text{Cov}(X, X) = \text{Var}(X)$ .

**Lemma 42** (Variance of a sum).  $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$ . *If  $X, Y$  are independent then  $\text{Cov}(X, Y) = 0$  and the variances simply add.*

*Proof.* Let  $D_X := X - \mu_X, D_Y := Y - \mu_Y$ . Then  $X+Y - (\mu_X + \mu_Y) = D_X + D_Y$ , and  $(D_X + D_Y)^2 = D_X^2 + 2D_X D_Y + D_Y^2$ . Take expectations, apply lemma 34, and read off the three terms. For the second claim: under independence  $\mathbb{E}[D_X D_Y] = \mathbb{E}[D_X]\mathbb{E}[D_Y] = 0 \cdot 0 = 0$ .  $\square$

*Remark 43* (The converse is false, and it matters here).  $\text{Cov}(X, Y) = 0$  does *not* imply independence. Covariance detects only linear co-movement. A quantity can be completely determined by another and still have zero covariance with it. Every “uncorrelated, therefore unrelated” inference in this literature is invalid for this reason, and §16 uses a *non-linear* measure of dependence precisely to avoid it.

**Theorem 44** (Cauchy–Schwarz).  $|\text{Cov}(X, Y)| \leq \text{sd}(X) \text{sd}(Y)$ .

*Proof.* Write  $D_X, D_Y$  as above and let  $t \in \mathbb{R}$ . Since a square is never negative,

$$0 \leq \mathbb{E}[(D_X + tD_Y)^2] = \mathbb{E}[D_X^2] + 2t \mathbb{E}[D_X D_Y] + t^2 \mathbb{E}[D_Y^2] = \text{Var}(X) + 2t \text{Cov}(X, Y) + t^2 \text{Var}(Y).$$

The right-hand side is a quadratic in  $t$  that is never negative. A quadratic  $at^2 + bt + c$  with  $a \geq 0$  is non-negative for all  $t$  if and only if its discriminant  $b^2 - 4ac \leq 0$  (otherwise it has two distinct real roots and dips below zero between them). Here  $a = \text{Var}(Y)$ ,  $b = 2 \text{Cov}(X, Y)$ ,  $c = \text{Var}(X)$ , so

$$4 \text{Cov}(X, Y)^2 - 4 \text{Var}(X) \text{Var}(Y) \leq 0 \implies \text{Cov}(X, Y)^2 \leq \text{Var}(X) \text{Var}(Y).$$

Take square roots.  $\square$

**Definition 45** (Correlation). If both standard deviations are non-zero,

$$\text{Corr}(X, Y) = r := \frac{\text{Cov}(X, Y)}{\text{sd}(X) \text{sd}(Y)}.$$

**Corollary 46.**  $-1 \leq r \leq 1$ , immediately from theorem 44.

So  $r$  is covariance with the units divided out. It is *not* a percentage, it is *not* a share of anything, and  $r = 0.5$  is not “half”. The only quantity with a share interpretation is  $r^2$ , and only under a linear model.

### 3.6 Samples, estimators, and the thing that is actually random

We never see  $\mathbb{P}$ . We see  $n$  observations  $x_1, \dots, x_n$  and must guess properties of  $\mathbb{P}$  from them.

**Definition 47** (Estimator). An *estimator* is any function of the observed data used to guess a fixed unknown quantity (the *estimand*). We write  $\hat{\theta}$  for an estimator of  $\theta$ .

**Definition 48** (Bias).  $\text{Bias}(\hat{\theta}) := \mathbb{E}[\hat{\theta}] - \theta$ . An estimator is *unbiased* if this is zero: it is right *on average across repetitions of the whole study*, which says nothing about any single study.

*Remark 49* (What is random).  $\theta$  is a fixed number.  $\hat{\theta}$  varies because the sample varies. Every statement below about “the distribution of  $\hat{\theta}$ ” is a statement about what would happen across hypothetical repetitions of the study, not about  $\theta$ . This is the single most misread point in applied statistics and it is the reason §3.10 states explicitly what a confidence interval does *not* mean.

### 3.7 The sample mean, and the derivation of the standard error

Let  $X_1, \dots, X_n$  be independent, each with mean  $\mu$  and variance  $\sigma^2$ . Define  $\bar{X} := \frac{1}{n} \sum_i X_i$ .

*Sub-step 0.7.1 — the mean is unbiased.*

By lemma 34,  $\mathbb{E}[\bar{X}] = \frac{1}{n} \sum_i \mathbb{E}[X_i] = \frac{1}{n} \cdot n\mu = \mu$ .

*Sub-step 0.7.2 — the variance of the mean.*

By lemma 39 with  $a = 1/n$ , then lemma 42 with independence:

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \text{Var}\left(\sum_i X_i\right) = \frac{1}{n^2} \sum_i \text{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}. \quad (3)$$

*Sub-step 0.7.3 — naming it.*

**Definition 50** (Standard error).  $\text{se}(\bar{X}) := \text{sd}(\bar{X}) = \sigma/\sqrt{n}$ .

*Remark 51* (The  $\sqrt{n}$  is the whole economics of measurement). Halving the standard error requires *four times* the data. This single fact determines every sample-size statement in §3.12, and it is why the question “how many prompts do we need” has an answer that is often unaffordable.

*Remark 52* (Independence is doing real work in equation (3)). If the  $X_i$  are positively correlated, lemma 42 adds covariance terms and the true variance is *larger* than  $\sigma^2/n$ . Using equation (3) anyway understates the uncertainty. §3.13 quantifies exactly how much, because this specific error occurred in this work.

### 3.8 Why $n - 1$

To use  $\text{se} = \sigma/\sqrt{n}$  we need  $\sigma^2$ , which we also do not know. The obvious estimator  $\frac{1}{n} \sum_i (X_i - \bar{X})^2$  is biased downward, and the reason is worth the derivation because it exposes what “degrees of freedom” means.

**Lemma 53.**  $\mathbb{E}\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] = (n - 1)\sigma^2$ .

*Proof.* Expand around the true mean  $\mu$  by adding and subtracting it:

$$\sum_i (X_i - \bar{X})^2 = \sum_i ((X_i - \mu) - (\bar{X} - \mu))^2 = \sum_i (X_i - \mu)^2 - 2(\bar{X} - \mu) \sum_i (X_i - \mu) + n(\bar{X} - \mu)^2.$$

Now  $\sum_i (X_i - \mu) = n(\bar{X} - \mu)$ , so the middle term is  $-2n(\bar{X} - \mu)^2$  and the last two combine to  $-n(\bar{X} - \mu)^2$ :

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Take expectations. The first term is  $n\sigma^2$ . For the second,  $\mathbb{E}[(\bar{X} - \mu)^2] = \text{Var}(\bar{X}) = \sigma^2/n$  by equation (3), so it contributes  $n \cdot \sigma^2/n = \sigma^2$ . Hence the expectation is  $n\sigma^2 - \sigma^2 = (n - 1)\sigma^2$ .  $\square$

**Definition 54** (Sample variance).  $s^2 := \frac{1}{n - 1} \sum_i (X_i - \bar{X})^2$ , which by lemma 53 satisfies  $\mathbb{E}[s^2] = \sigma^2$ .

*The interpretation.*

Deviations are taken from  $\bar{X}$ , which was computed from the same data and therefore sits closer to the points than  $\mu$  does. One degree of freedom was spent estimating the centre. Dividing by  $n - 1$  returns it.

### 3.9 The central limit theorem, and precisely what it licenses

**Theorem 55** (Central limit theorem, stated). *If  $X_1, \dots, X_n$  are independent and identically distributed with mean  $\mu$  and finite variance  $\sigma^2$ , then as  $n$  grows the distribution of  $(\bar{X} - \mu)/(\sigma/\sqrt{n})$  approaches the standard normal distribution:  $\mathbb{P}(Z \leq z) \rightarrow \Phi(z)$  where  $\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$ .*

We do not prove this. What matters is what it does and does not say.

- **Says:** the *average* becomes normally distributed regardless of the shape of the individual observations. This is why a rate built from 0/1 indicators – about as non-normal as data gets – still has a normal sampling distribution.
- **Does not say:** that the data are normal. They are not, and it does not matter.
- **Fails when:** observations are dependent (§3.13), or  $n$  is small with a skewed distribution, or the quantity is not an average. In this paper the third case arises for correlations, which is why §3.14 uses resampling instead of [theorem 55](#) there.

Three quantiles of  $\Phi$  are used throughout and are quoted once here:

$$z_{0.975} = 1.9600, \quad z_{0.80} = 0.8416, \quad z_{0.99} = 2.3263. \quad (4)$$

$z_q$  is defined by  $\Phi(z_q) = q$ : the point with  $q$  of the distribution below it.

### 3.10 Confidence intervals: derivation, and what they do not mean

*Sub-step 0.10.1.*

By [theorem 55](#), with probability 0.95,

$$-1.96 \leq \frac{\bar{X} - \mu}{\text{se}} \leq 1.96.$$

*Sub-step 0.10.2.*

Multiply through by  $\text{se}$  (positive, so inequalities are preserved):  $-1.96 \text{ se} \leq \bar{X} - \mu \leq 1.96 \text{ se}$ .

*Sub-step 0.10.3.*

Subtract  $\bar{X}$  and multiply by  $-1$ , which *reverses* both inequalities:  $\bar{X} - 1.96 \text{ se} \leq \mu \leq \bar{X} + 1.96 \text{ se}$ .

**Definition 56** (95% confidence interval).  $[\bar{X} - 1.96 \text{ se}, \bar{X} + 1.96 \text{ se}]$ .

*Remark 57* (What it means). Across hypothetical repetitions of the entire study, 95% of the intervals so constructed would contain the fixed unknown  $\mu$ . The randomness is in the interval, not in  $\mu$ .

*Remark 58* (What it does not mean). It does *not* mean  $\mathbb{P}(\mu \in [\ell, u]) = 0.95$  for the particular interval computed:  $\mu$  is a constant and either lies in it or does not. It does *not* mean 95% of future observations fall in it – that is a prediction interval and is much wider. And an interval that excludes zero does not mean the effect is large; §3.12 separates those two questions.

### 3.11 Hypothesis tests, $z$ -statistics, and $p$ -values

**Definition 59** (Null hypothesis). A fully specified statement about the estimand, denoted  $H_0$ , typically  $\theta = 0$ . It is specified in advance so that the distribution of the estimator *under* it is known.

**Definition 60** ( $z$ -statistic).  $z := \hat{\theta}/\text{se}(\hat{\theta})$ : the estimate measured in units of its own uncertainty. A  $z$  of 3 means the observed effect is three standard errors from zero.

**Definition 61** ( $p$ -value).  $p := \mathbb{P}(|Z| \geq |z| \mid H_0)$ , the probability of seeing an effect at least this extreme *if the null were true*.

*Remark 62* (The three standard misreadings, each false).  $p$  is not  $\mathbb{P}(H_0 \mid \text{data})$  – it is conditioned the other way round.  $p$  is not the probability the result is a fluke. And  $1 - p$  is not the probability the effect is real. The  $p$ -value is a statement about data under an assumption, never about the assumption given data.

*Remark 63* (Why this paper mostly reports  $z$  and intervals). A  $p$ -value collapses effect size and precision into one number, so a tiny effect measured precisely and a large effect measured sloppily can produce the same  $p$ . The interval keeps them separate, which is why every finding in §18 carries one.

### 3.12 Power, and the minimum detectable effect

A null result is uninterpretable without knowing what the design could have seen.

**Definition 64** (Power). The probability of rejecting  $H_0$  when a specific alternative  $\theta = \delta$  is true. Conventionally  $1 - \beta = 0.80$ .

**Theorem 65** (Minimum detectable effect). *For a two-sided test at level  $\alpha$  with power  $1 - \beta$ , the smallest true effect detectable with that power is*

$$\text{MDE} = (z_{1-\alpha/2} + z_{1-\beta}) \cdot \text{se}.$$

*Proof. Sub-step 1.* We reject when  $|\hat{\theta}| \geq z_{1-\alpha/2} \text{se}$ , by construction of the level- $\alpha$  test.

*Sub-step 2.* Suppose the truth is  $\theta = \delta > 0$ . Then  $\hat{\theta}$  is centred at  $\delta$  with standard deviation  $\text{se}$ , so  $(\hat{\theta} - \delta)/\text{se}$  is standard normal.

*Sub-step 3.* Rejection (ignoring the negligible wrong-tail probability) requires  $\hat{\theta} \geq z_{1-\alpha/2} \text{se}$ , i.e.

$$\frac{\hat{\theta} - \delta}{\text{se}} \geq z_{1-\alpha/2} - \frac{\delta}{\text{se}}.$$

*Sub-step 4.* That event has probability  $1 - \beta$  exactly when the threshold on the right equals  $-z_{1-\beta}$ , because  $\mathbb{P}(Z \geq -z_{1-\beta}) = 1 - \beta$  by symmetry of the normal.

*Sub-step 5.* So  $z_{1-\alpha/2} - \delta/\text{se} = -z_{1-\beta}$ , giving  $\delta = (z_{1-\alpha/2} + z_{1-\beta}) \text{se}$ .  $\square$

**Corollary 66** (The number used throughout). At  $\alpha = 0.05$ ,  $1 - \beta = 0.80$ : from [equation \(4\)](#),  $\text{MDE} = (1.9600 + 0.8416) \text{se} = 2.8016 \text{se}$ .

**Corollary 67** (Required sample size). Since  $\text{se} = \sigma/\sqrt{n}$ , detecting an effect of size  $\delta$  needs  $n = (2.8016 \sigma/\delta)^2$ .

*Remark 68* (Reading a null). If  $\text{MDE} > |\text{the effect the null was supposed to rule out}|$ , the null is *silence*, not evidence of absence. §18 reports an MDE beside every null for this reason.

### 3.13 Clustering, the intraclass correlation, and the design effect

This subsection exists because the error it describes was made in this work and had to be corrected.

*The setting.*

Suppose the data have two levels:  $P$  independent *clusters* (here: prompts), and within each cluster  $R$  observations (here: repeated perturbations of the same prompt). Observations in the same cluster are *not* independent.

*Sub-step 0.13.1 — the model.*

Write  $X_{pr} = \mu + a_p + e_{pr}$ , where  $a_p$  is a cluster effect with variance  $\sigma_b^2$  (“between”),  $e_{pr}$  is an observation effect with variance  $\sigma_w^2$  (“within”), and all terms are independent of each other.

*Sub-step 0.13.2 — the cluster mean.*

$\bar{X}_p = \mu + a_p + \bar{e}_p$ , and by [lemma 42](#) and [equation \(3\)](#),  $\text{Var}(\bar{X}_p) = \sigma_b^2 + \sigma_w^2/R$ .

*Sub-step 0.13.3 — the overall mean.*

The  $P$  cluster means *are* independent, so [equation \(3\)](#) applies to them:

$$\text{Var}(\hat{\pi}) = \frac{1}{P} \left( \sigma_b^2 + \frac{\sigma_w^2}{R} \right). \quad (5)$$

*Sub-step 0.13.4 — what the naive formula would give.*

Treating all  $PR$  observations as independent with total variance  $\sigma^2 = \sigma_b^2 + \sigma_w^2$  gives  $\sigma^2/(PR)$ .

**Definition 69** (Intraclass correlation).  $\rho := \frac{\sigma_b^2}{\sigma_b^2 + \sigma_w^2}$ , the share of total variance that is between clusters. Equivalently, the correlation between two observations drawn from the same cluster.

**Theorem 70** (Design effect).

$$\text{DEFF} := \frac{\text{correct variance } \text{equation (5)}}{\text{naive variance}} = 1 + (R - 1)\rho.$$



*Proof.* Divide equation (5) by  $\sigma^2/(PR)$ :

$$\frac{\frac{1}{P}(\sigma_b^2 + \sigma_w^2/R)}{\frac{\sigma_b^2 + \sigma_w^2}{PR}} = \frac{R\sigma_b^2 + \sigma_w^2}{\sigma_b^2 + \sigma_w^2} = \frac{R\sigma_b^2 + \sigma_w^2 + (\sigma_b^2 - \sigma_b^2)}{\sigma_b^2 + \sigma_w^2} = 1 + \frac{(R-1)\sigma_b^2}{\sigma_b^2 + \sigma_w^2} = 1 + (R-1)\rho. \quad \square$$

**Corollary 71.** *Standard errors computed as if the PR observations were independent are too small by a factor  $\sqrt{1 + (R-1)\rho}$ , and z-statistics are too large by the same factor.*

*Remark 72* (Effective sample size).  $n_{\text{eff}} = PR/\text{DEFF}$ . With  $R = 20$  and  $\rho = 0.27$ ,  $\text{DEFF} = 6.13$ , so 19,360 rows carry the information of about 3,150 – and the honest count of independent units remains  $P = 968$ .

### 3.14 The bootstrap

The formulas above give se for a *mean*. For a correlation, a median, a ratio, or anything else, no such formula is available in general. The bootstrap replaces it with computation.

**Protocol 73** (Cluster bootstrap). Given clusters  $c_1, \dots, c_P$  and a statistic  $T$ :

1. Draw  $P$  clusters *with replacement* from  $\{c_1, \dots, c_P\}$ .
2. Compute  $T$  on the resampled collection; call it  $T^{(1)}$ .
3. Repeat  $B$  times, obtaining  $T^{(1)}, \dots, T^{(B)}$ .
4. Report the 2.5th and 97.5th percentiles as a 95% interval.

*Why it works, in one line.*

The variation of  $T$  across resamples of the observed data mimics the variation of  $T$  across fresh samples from the population, because the observed sample is our best available stand-in for that population.

*Why the resampling unit must be the cluster.*

Resampling *rows* would create datasets whose within-cluster dependence differs from the real one, and would reproduce the understatement of [theorem 70](#). Every interval in this paper resamples prompts.

*When it fails.*

With few clusters, at the boundary of a parameter's range, or for statistics that are not smooth functions of the data. All three are checked in §22.

### 3.15 Permutation tests, and exactly what they test

**Protocol 74** (Permutation test). To test whether the *pairing* between two vectors  $x$  and  $y$  carries information: compute  $T(x, y)$ ; then repeatedly shuffle  $x$  and recompute  $T(x_{\text{shuffled}}, y)$ ; the  $p$ -value is the fraction of shuffles reaching  $|T|$  at least as large as observed.

*Remark 75* (The exchangeability assumption). Shuffling assumes that under  $H_0$  all orderings are equally likely. If  $x$  has internal structure – clustering, time order – shuffling destroys that too, and the test then answers a question nobody asked. In §18 the shuffle is applied to cluster-level values for this reason.

*Remark 76* (Resolution floor). With  $B$  shuffles the smallest attainable  $p$  is  $1/(B+1)$ : a reported  $p = 0.0000$  from  $B = 200$  means only  $p < 0.005$ , and no more.

*Remark 77* (What a permutation null cannot do). It answers *did the pairing matter*, never *why*. It is a falsifier, never a confirmer of a mechanism.

### 3.16 Multiplicity

**Theorem 78** (Bonferroni). *If  $C$  tests are each performed at level  $\alpha/C$ , then the probability of at least one false rejection when all nulls are true is at most  $\alpha$ .*

*Proof.* Let  $A_i$  be the event of a false rejection on test  $i$ , so  $\mathbb{P}(A_i) = \alpha/C$ . The union bound gives  $\mathbb{P}(\bigcup_i A_i) \leq \sum_i \mathbb{P}(A_i) = C \cdot \alpha/C = \alpha$ . The union bound itself follows from [corollary 35](#) plus  $\mathbf{1}\{\bigcup_i A_i\} \leq \sum_i \mathbf{1}\{A_i\}$ , which holds pointwise because the left side is 0 or 1 and the right side counts.  $\square$

**Corollary 79** (The bar used in this paper). *For  $C \approx 170$  tests at  $\alpha = 0.05$ :  $\alpha/C \approx 0.0003$ , whose two-sided normal quantile is  $|z| \approx 3.6$ . This paper uses the more conservative  $|z| > 3.9$  and reports the number of tests and the number of survivors together, since reporting only survivors is the multiplicity failure with manners.*



*Remark 80* (Bonferroni versus false discovery rate). Bonferroni controls the probability of *any* false positive, which is the right target when a single false claim is costly. The Benjamini–Hochberg procedure instead controls the expected *share* of false positives among rejections, and rejects hypothesis  $(k)$  when  $p_{(k)} \leq qk/C$  for the largest such  $k$ . Its threshold at rank  $k = C$  is  $q$  itself, not  $q/C$  – confusing the two demands orders of magnitude more resolution than BH actually requires.

### 3.17 Measurement error and attenuation

Every quantity in §5 is measured by a fallible instrument. This subsection derives the exact consequence.

*Sub-step 0.17.1 — the model.*

Suppose the truth is  $T$  but we observe  $X = T + \varepsilon$ , with  $\varepsilon$  independent of  $T$ ,  $\mathbb{E}[\varepsilon] = 0$ ,  $\text{Var}(\varepsilon) = \sigma_\varepsilon^2$ .

*Sub-step 0.17.2 — reliability.*

Define

$$\text{rel} := \frac{\text{Var}(T)}{\text{Var}(X)} = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\varepsilon^2} \in [0, 1],$$

the share of observed variance that is signal. [Lemma 42](#) with independence gives the denominator.

*Sub-step 0.17.3 — two independent measurements.*

If  $X_1 = T + \varepsilon_1$  and  $X_2 = T + \varepsilon_2$  with  $\varepsilon_1 \perp \varepsilon_2$  and equal error variances, then  $\text{Cov}(X_1, X_2) = \text{Var}(T) = \sigma_T^2$  (the error terms contribute nothing), and each has variance  $\sigma_T^2 + \sigma_\varepsilon^2$ . Therefore

$$\text{Corr}(X_1, X_2) = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\varepsilon^2} = \text{rel}. \quad (6)$$

The correlation between two independent measurements of the same thing is the reliability of one of them. This is what makes §10 computable at zero extra cost.

**Theorem 81** (Attenuation). *Let  $X = T + \varepsilon_X$  and  $Y = U + \varepsilon_Y$  with independent errors. Then*

$$\text{Corr}(X, Y) = \text{Corr}(T, U) \cdot \sqrt{\text{rel}_X \text{rel}_Y}.$$

*Proof.*  $\text{Cov}(X, Y) = \text{Cov}(T + \varepsilon_X, U + \varepsilon_Y) = \text{Cov}(T, U)$ , since all three cross terms involve an error independent of everything else. The denominators are  $\text{sd}(X) = \sigma_T / \sqrt{\text{rel}_X}$  and likewise for  $Y$ , because  $\text{rel}_X = \sigma_T^2 / \sigma_X^2$  rearranges to  $\sigma_X = \sigma_T / \sqrt{\text{rel}_X}$ . Substituting,

$$\text{Corr}(X, Y) = \frac{\text{Cov}(T, U)}{\frac{\sigma_T}{\sqrt{\text{rel}_X}} \cdot \frac{\sigma_U}{\sqrt{\text{rel}_Y}}} = \frac{\text{Cov}(T, U)}{\sigma_T \sigma_U} \sqrt{\text{rel}_X \text{rel}_Y}. \quad \square$$

**Corollary 82** (The ceiling). *A perfect true relationship,  $\text{Corr}(T, U) = 1$ , reads at most  $\sqrt{\text{rel}_X \text{rel}_Y}$ . No amount of data raises this: it is bias, not noise, and  $\sqrt{n}$  does not touch it.*

### 3.18 Reliability of an aggregate, and the Spearman–Brown formula

Averaging several noisy measurements reduces error. The exact statement:

**Theorem 83** (Spearman–Brown). *If a score is the sum of  $m$  exchangeable components each with reliability correlation  $r$  to another such component, then doubling the number of components gives reliability  $2r/(1+r)$ .*

*Derivation.* Let each component be  $T + \varepsilon_i$  with  $\text{Var}(T) = \sigma_T^2$ ,  $\text{Var}(\varepsilon_i) = \sigma_\varepsilon^2$ , errors independent. The sum of  $m$  components is  $mT + \sum_i \varepsilon_i$ , with signal variance  $m^2 \sigma_T^2$  and error variance  $m \sigma_\varepsilon^2$  by [lemma 42](#). So

$$\text{rel}_m = \frac{m^2 \sigma_T^2}{m^2 \sigma_T^2 + m \sigma_\varepsilon^2} = \frac{m \sigma_T^2}{m \sigma_T^2 + \sigma_\varepsilon^2} = \frac{m \lambda}{m \lambda + 1}, \quad \lambda := \sigma_T^2 / \sigma_\varepsilon^2.$$

For  $m = 1$  this is  $r = \lambda / (\lambda + 1)$ , so  $\lambda = r / (1 - r)$ . Substituting  $m = 2$ :

$$\text{rel}_2 = \frac{2r/(1-r)}{2r/(1-r) + 1} = \frac{2r}{2r + (1-r)} = \frac{2r}{1+r}. \quad \square$$

*Remark 84* (Use here). Splitting the annotators of a prompt into two halves and correlating the two half-scores estimates the reliability of a *half*-sized panel. Applying [theorem 83](#) corrects it to the full panel, which is the quantity that actually enters [theorem 81](#).

### 3.19 Two-way clustering

When observations are grouped along two crossed dimensions at once – here, prompts and annotators, since an annotator appears on several prompts and a prompt is rated by several annotators – neither one-way correction suffices.

**Theorem 85** (Cameron–Gelbach–Miller).  $\hat{V}_{two-way} = \hat{V}_{prompt} + \hat{V}_{annotator} - \hat{V}_{prompt \times annotator}$ .

*Sketch.* Two observations are dependent if they share a prompt *or* an annotator. By inclusion–exclusion applied to the covariance terms of lemma 42, pairs sharing *both* are counted once in each of the first two sums and must be subtracted once. The estimator can be slightly negative in finite samples, which signals that one dimension carries almost no dependence.  $\square$

## 4 Every tool, executed by hand

Part 0 derived the machinery. This section runs all of it on five numbers, so that nothing is taken on trust. Take the sample

$$x = (2, 4, 4, 4, 6), \quad n = 5.$$

### 4.1 The mean

$$\bar{x} = \frac{2 + 4 + 4 + 4 + 6}{5} = \frac{20}{5} = 4.$$

### 4.2 Deviations, and why the naive spread is useless

$$x_i - \bar{x} = (-2, 0, 0, 0, 2), \quad \sum_i (x_i - \bar{x}) = 0.$$

Exactly zero, as §0.4 said it must be for *any* dataset. This is why the average deviation cannot be a measure of spread.

### 4.3 Variance, both ways

Squared deviations: (4, 0, 0, 0, 4), summing to 8.

$$s^2 = \frac{8}{n-1} = \frac{8}{4} = 2, \quad s = \sqrt{2} = 1.4142.$$

Check with lemma 37.  $\mathbb{E}[X^2]$  over the sample is  $(4 + 16 + 16 + 16 + 36)/5 = 88/5 = 17.6$ , and  $\bar{x}^2 = 16$ , so the *population-form* variance is  $17.6 - 16 = 1.6 = 8/5$ . Multiplying by  $n/(n-1) = 5/4$  recovers 2. The two forms agree, and the factor between them is exactly lemma 53.

### 4.4 Standard error and interval

$$se = \frac{s}{\sqrt{n}} = \frac{1.4142}{2.2361} = 0.6325.$$

By §3.10 the 95% interval is  $4 \pm 1.96(0.6325) = 4 \pm 1.2397 = [2.760, 5.240]$ .

*Read it correctly.*

Across repetitions of the whole study, 95% of such intervals cover the fixed unknown  $\mu$ . It does *not* say 95% of future observations land in  $[2.760, 5.240]$ ; three of the five observed values are exactly 4, and a prediction interval would be far wider.

### 4.5 A $z$ -statistic and its $p$ -value

Testing  $H_0 : \mu = 0$ :  $z = 4/0.6325 = 6.32$ . From the normal table,  $\mathbb{P}(|Z| \geq 6.32) < 10^{-9}$ .

*Read it correctly.*

This is  $\mathbb{P}(\text{data this extreme} \mid \mu = 0)$ . It is not the probability that  $\mu = 0$ , and  $1 - p$  is not the probability the effect is real.

### 4.6 Minimum detectable effect, and required $n$

By theorem 65,  $MDE = 2.8016 \times 0.6325 = 1.772$ : with five observations of this spread, a true mean below 1.772 would be missed more often than not. By corollary 67, detecting  $\delta = 0.5$  needs

$$n = \left( \frac{2.8016 \times 1.4142}{0.5} \right)^2 = (7.923)^2 = 62.8 \rightarrow 63.$$

Detecting  $\delta = 0.25$  needs  $4\times$  that: 251. This is the  $\sqrt{n}$  economics of §3.7 in one line.

#### 4.7 An indicator, a proportion, and its variance

Let  $P_i$  be “ $x_i > 3$ ”, giving indicators  $(0, 1, 1, 1, 1)$ . By [equation \(1\)](#) the proportion is  $4/5 = 0.8$ ; by [corollary 38](#) its variance is  $0.8 \times 0.2 = 0.16$ , so  $se = \sqrt{0.16/5} = \sqrt{0.032} = 0.1789$ .

*Why rates near the ends are measured better.*

$\pi(1 - \pi)$  peaks at  $\pi = 0.5$  with value 0.25 and falls to 0.09 at  $\pi = 0.1$ . A 5% flip rate is therefore intrinsically easier to pin down than a 50% one, which is why the intervals in [table 11](#) are narrow despite modest  $n$ .

#### 4.8 Covariance and correlation

Let  $y = (1, 3, 3, 5, 3)$ , so  $\bar{y} = 3$  and  $y_i - \bar{y} = (-2, 0, 0, 2, 0)$ .

$$\text{Cov} = \frac{1}{n-1} \sum_i (x_i - \bar{x})(y_i - \bar{y}) = \frac{(-2)(-2) + 0 + 0 + 0 \cdot 2 + 2 \cdot 0}{4} = \frac{4}{4} = 1.$$

$s_y^2 = (4 + 0 + 0 + 4 + 0)/4 = 2$ , so  $s_y = 1.4142$ , and

$$r = \frac{1}{1.4142 \times 1.4142} = \frac{1}{2} = 0.5.$$

[Theorem 44](#) requires  $|r| \leq 1$ : satisfied. And  $r = 0.5$  is *not* “half of anything”; only  $r^2 = 0.25$  has a share reading, and only under a linear model.

#### 4.9 The design effect, by hand

Suppose  $P = 3$  clusters of  $R = 4$ :

$$c_1 = (1, 1, 1, 0), \quad c_2 = (0, 0, 0, 0), \quad c_3 = (1, 0, 1, 1).$$

Cluster means: 0.75, 0, 0.75. Grand mean = 0.5.

$$\begin{aligned} \sigma_b^2 &= \frac{1}{2} [(0.25)^2 + (0.5)^2 + (0.25)^2] = \frac{1}{2}(0.375) = 0.1875, \\ \sigma_w^2 &= \text{mean of within-cluster variances} = \frac{1}{3} [0.25 + 0 + 0.25] = 0.1667, \end{aligned}$$

using  $s^2 = \frac{1}{R-1} \sum (x - \bar{x})^2$  inside each cluster.

$$\rho = \frac{0.1875}{0.1875 + 0.1667} = 0.529, \quad \text{DEFF} = 1 + (4 - 1)(0.529) = 2.59.$$

So treating the 12 rows as independent would understate the standard error by  $\sqrt{2.59} = 1.61$ , and the honest count of independent units is 3, not 12.

#### 4.10 Attenuation, by hand

Suppose a true correlation  $\text{Corr}(T, U) = 0.90$ , measured through instruments with reliabilities 0.55 and 0.95. By [theorem 81](#)

$$\text{Corr}(X, Y) = 0.90 \times \sqrt{0.55 \times 0.95} = 0.90 \times \sqrt{0.5225} = 0.90 \times 0.7228 = 0.6505.$$

A strong relationship reads as a moderate one, and no sample size repairs it: the attenuation is bias, and  $se$  shrinks while the centre stays wrong.

#### 4.11 Spearman–Brown, by hand

A half-panel split-half correlation of  $r = 0.8961$  gives, by [theorem 83](#),

$$\frac{2(0.8961)}{1 + 0.8961} = \frac{1.7922}{1.8961} = 0.9452,$$

which is the reliability of the *full* panel and the number that enters [theorem 81](#).

#### 4.12 Bonferroni, by hand

With  $C = 171$  tests at  $\alpha = 0.05$ : per-test level  $0.05/171 = 0.000292$ . The two-sided normal quantile at  $1 - 0.000292/2 = 0.999854$  is  $|z| = 3.62$ . This paper uses 3.9, which is stricter, and reports the count of tests beside the count of survivors.

## 5 The object

### 5.1 What a normative disposition is, formally

**Definition 86** (Disposition). Let  $\mathcal{X}$  be the set of all possible responses to a prompt – unbounded. A person  $i$ ’s *normative disposition* is a preference structure  $\succeq_i$  on  $\mathcal{X}$ . It is not a number, not a sentence, and not a ranking: it is a relation on an infinite set.

**Definition 87** (Elicitation). An *elicitation* observes  $\succeq_i$  only on a finite *probe set*  $X = \{x_1, \dots, x_k\} \subset \mathcal{X}$ . In the object studied here,  $k = 4$ .

### 5.2 The chain

$$\succeq_i \rightarrow N_i \rightarrow A \rightarrow G \rightarrow C \rightarrow J \rightarrow D \rightarrow Y \quad (7)$$

$N_i$  the elicited individual norm;  $A$  the aggregation rule;  $G = A(N_1, \dots, N_n)$  the collective norm;  $C$  the compiled standard;  $J$  the executor that reads it;  $D$  the decision it induces;  $Y$  the behaviour of a system trained under it.

**Remark 88** (Three kinds of loss, which must never be merged). 1. **Mathematically forced:** no pipeline can avoid it (theorem 94).

2. **Authorised social choice:** aggregation necessarily discards individual information (theorem 104). This is a *defect* only if the rule  $A$  was never declared.

3. **Machine distortion:** an unlicensed change.

The released object declares no  $A$ . *Therefore categories 2 and 3 are not separable at this site.* That is a missing column, not a hard measurement problem, and it is the single most consequential fact about the release.

### 5.3 The tensors, exactly

**Definition 89** (The released quantities). For prompt  $p$ , criterion index  $c$ , response letter  $r \in \{A, B, C, D\}$ , annotator  $a$ :

|                                                    |                                                 |
|----------------------------------------------------|-------------------------------------------------|
| $S[p, c, r] \in (0, 1)$                            | satisfaction of criterion $c$ by response $r$ ; |
| $W[p, c, a] \in [-10, 10] \setminus \{0\}$         | the weight $a$ gave $c$ ;                       |
| $V[p, a] \subseteq \{A, B, C, D\}$                 | the responses $a$ called unacceptable;          |
| $\Pi[p, a] \in \text{rankings of } \{A, B, C, D\}$ | the ranking $a$ gave.                           |

Four facts about definition 89 are load-bearing and are stated here so that no later section can assume otherwise.

1.  $S$  is *not* in the release. It is reconstructed by running a language model as a judge; the value is a sigmoid of a logit difference and has never been calibrated against anything. Every number downstream is a number about that judge as much as about the norm (theorem 81, §10).
2.  $c$  is *ragged*: between 4 and 39 criteria per prompt, median 15. Any statistic that averages over criteria therefore weights prompts unequally unless it is explicitly re-weighted.
3.  $W$  is *sparse*: 102,147 observed  $(c, a)$  pairs. Exactly one rating in the corpus equals 0, so the scale is used as strictly two-sided.
4.  $V$  and  $W$  **do not share an index**.  $V$  is indexed by (prompt, annotator);  $W$  by (prompt, criterion, annotator). There is no column linking a veto to a criterion. §8 shows what this forbids.

**Remark 90** (The index itself is reconstructed). The rubric file carries no prompt identifier. The mapping from criteria to prompts is recovered by matching conversation text, with a fuzzy fallback: 966 exact, 2 fuzzy, and 18 of 986 records **never matched at all**. So 1.8% of the rubric corpus is invisible to every number computed from this release, including every number in this paper.

### 5.4 What a claim about this object can mean

**Definition 91** (Induced score).  $Y_p(r) := \sum_c \bar{w}_{p,c} S[p, c, r]$ , where  $\bar{w}_{p,c}$  is the mean over annotators of  $W[p, c, a]$ .

**Definition 92** (Individual induced score).  $N_{i,p}(r) := \sum_{c: i \text{ rated } c} W[p, c, i] S[p, c, r]$ .

**Remark 93** (Why  $N_i$  must be centred and normalised, and why that is forced). Only contrasts among the four responses decide anything, so adding a constant to  $N_i$  changes nothing; the mean over  $r$  is therefore unidentified and must be removed. Furthermore,  $\|N_i\|$  grows mechanically with the number of criteria  $i$  rated:  $\text{measured Corr}(\log \|N_i\|, \log \#\{\text{criteria rated}\}) = +0.842$ . Any statistic on unnormalised  $N_i$  therefore measures *effort*, not values. Centring and normalising are not stylistic; without them the quantity is a different quantity.

## 6 The elicitation layer

### 6.1 The probe bound

**Theorem 94** (Probe bound). *Let  $u : X \rightarrow \mathbb{R}$  be a utility on  $k$  probe responses. If decisions depend only on differences of  $u$ , the space of decision-distinct utilities has dimension  $k - 1$ . If in addition only the ordering matters (no interpersonal scale), the space is  $S^{k-2}$ , of dimension  $k - 2$ .*

*Proof. Step 1.* The set of utilities is  $\mathbb{R}^k$ .

*Step 2 (quotient by constants).*  $u$  and  $u + c\mathbf{1}$  induce identical differences  $u(x_a) - u(x_b)$  for every pair, hence identical decisions. The subspace of constants  $\mathbb{R}\mathbf{1}$  has dimension 1, so the quotient  $\mathbb{R}^k / \mathbb{R}\mathbf{1}$  has dimension  $k - 1$ . Concretely, the centring map  $C = I - \frac{1}{k}\mathbf{1}\mathbf{1}^\top$  is a projection with  $\text{rank}(C) = k - 1$ ; for  $k = 4$  this rank is verified numerically as 3.

*Step 3 (quotient by scale).* If  $u$  and  $\lambda u$  for  $\lambda > 0$  are also equivalent, we quotient the  $(k - 1)$ -dimensional centred space by the positive rays, giving the sphere  $S^{k-2}$ , of dimension  $k - 2$ .

*Step 4.* For  $k = 4$ : dimension 3 after centring, and  $S^2$  – two free parameters – after scale.  $\square$

**Theory claim.** However many criteria a person writes and however many weights they attach, the elicitation returns **a point on a 2-sphere**: two free numbers per person per prompt. This is arithmetic, not a finding, and it is the ceiling on everything downstream.

**Corollary 95** (Three consequences). *1. The numerator of “how much normative information did the pipeline destroy” is bounded by two parameters. Most of what a criterion says never entered the measurable object at all.*  
*2. Any measure of loss based on rank or dimension must read a constant. An earlier round of this work built such a measure; it returned the value 3.00 on every input including its own null. That was not a bug – it was [theorem 94](#).*  
*3. Loss must be measured as angle and influence. You cannot lose a dimension you never had; you can only be rotated away from what people wanted.*

### 6.2 Is the weight scale ordinal or interval? A test, and an answer

Nothing in the elicitation establishes that the distance from +8 to +10 means the same as from +2 to +4. If the scale is merely *ordinal* – only the ordering of criteria within an annotator is meaningful – then any strictly increasing transformation of the weights must leave every decision unchanged.

**Protocol 96** (Monotone gauge test). Apply  $\varphi$  with  $\varphi$  strictly increasing and odd, recompute  $Y_p$  from [definition 91](#) with  $\varphi(\bar{w})$  in place of  $\bar{w}$ , and record whether  $\arg \max_r Y_p(r)$  moves.

| transform $\varphi(w)$         | winner changes |
|--------------------------------|----------------|
| identity (control)             | 0.0%           |
| $\text{sign}(w) w ^{1/2}$      | 4.4%           |
| $\text{sign}(w) \log(1 +  w )$ | 4.6%           |
| $\text{sign}(w) w ^2$          | 7.3%           |
| $\text{sign}(w)$ alone         | <b>11.7%</b>   |

**Table 2:** Every transform preserves each annotator’s ordering of criteria exactly. If the scale were ordinal, all rows would read 0.0%.  $n = 968$  prompts.

**Boundary.** The scale is *used* as an interval scale. 7.3% of decisions change under a transform that preserves every stated preference ordering, and 11.7% change when magnitude is discarded entirely. Every result in this literature that sums weighted criteria depends on an interval assumption the elicitation never established. The last row is the honest headline: roughly one decision in eight is carried by magnitude rather than direction.

### 6.3 What the elicitation captured: consensus, not the person

| quantity (14,764 person–prompt pairs)               | value                             |
|-----------------------------------------------------|-----------------------------------|
| $\cos(N_i, \text{ person } i\text{'s own ranking})$ | +0.3928                           |
| $\cos(N_i, \text{ a different person's ranking})$   | +0.3097                           |
| difference                                          | +0.0831 (se 0.0067, $z = +12.3$ ) |

**Table 3:** Both vectors centred and normalised per [remark 93](#).

Two readings, and the second is larger than the first.

- **Relative.** The entire individual signal is 0.083 of cosine. A person’s written criteria predict a *stranger’s* ranking almost as well as their own.
- **Absolute.** 0.393 means a person’s own stated criteria recover only about 39% of the direction of their own choice.

**Theory claim.** Before the pipeline does anything, roughly 61% of the relation between what a person *writes* and what they *choose* is already absent. No downstream step can destroy individual normative content that the elicitation never captured; “how much of a person survives compilation” is bounded above by how much of them was ever written down.

### 6.4 The normative payload actually collected

| field, by lexical marker                                      | criteria      | share        |
|---------------------------------------------------------------|---------------|--------------|
| exception / carve-out ( <i>unless, except, other than</i> )   | 123           | 0.8%         |
| conditional scope ( <i>if, when, where, in cases of</i> )     | 1,724         | 11.3%        |
| priority / conflict ( <i>before, over, takes precedence</i> ) | 440           | 2.9%         |
| uncertainty ( <i>may, might, unclear, depends</i> )           | 478           | 3.1%         |
| non-compensable ( <i>never, must not, regardless</i> )        | 121           | 0.8%         |
| <b>none of the five</b>                                       | <b>12,547</b> | <b>82.3%</b> |

**Table 4:** All 15,248 criteria. Median length 94 characters.

A minimal normative payload has been proposed with thirteen fields: content, scope, polarity, exceptions, priority, type, compensability, authority, judgment type, uncertainty, witnesses, constituency, meta-rules. This release populates **three**: content, polarity (the sign of a weight), magnitude.

**Corollary 97** (A question that is empty here). “*Did the compilation destroy the exception clause?*” *cannot be studied at this site: 99.2% of criteria contain no exception. A pipeline cannot destroy structure the elicitation never collected. This is a statement about the release, not about pipelines.*

## 7 The aggregation layer

### 7.1 Social choice, from zero

Given  $n$  people each with a ranking of  $k$  alternatives, a *social choice rule* outputs one alternative. Five rules appear below; each is defined completely.

**Definition 98** (Rules). Let  $\pi_i(x)$  be the Borda points person  $i$  gives  $x$  (i.e.  $k - 1$  for their first choice down to 0 for last, ties sharing the average).

- **Plurality:** pick the  $x$  that is ranked first by the most people.
- **Borda:** pick  $\arg \max_x \frac{1}{n} \sum_i \pi_i(x)$ .
- **Median:** pick  $\arg \max_x \text{median}_i \pi_i(x)$ .
- **Maximin:** pick  $\arg \max_x \min_i \pi_i(x)$  – the alternative whose *worst* treatment by any individual is least bad.
- **Condorcet winner:** the  $x$  that beats every other alternative in pairwise majority comparison. It need not exist.

**Definition 99** (Condorcet cycle). A majority cycle is a triple with  $x$  beating  $y$ ,  $y$  beating  $z$ , and  $z$  beating  $x$ . Majority preference is then not transitive and no Condorcet winner exists.

## 7.2 Measured social choice structure

| quantity (968 prompts, $\geq 3$ rankings each) | value       |
|------------------------------------------------|-------------|
| a Condorcet winner exists                      | 72.3%       |
| a majority <i>cycle</i> occurs                 | <b>0.0%</b> |
| neither (majority ties)                        | 27.7%       |

**Table 5:** The classic voting pathology does not occur in this corpus.

| agreement with the Condorcet winner, where one exists ( $n = 700$ ) |              |
|---------------------------------------------------------------------|--------------|
| Borda                                                               | 98.4%        |
| Plurality                                                           | 98.3%        |
| Median                                                              | 97.4%        |
| <b>the weighted-criteria score of definition 91</b>                 | <b>68.4%</b> |
| Maximin                                                             | 55.9%        |

**Theory claim.** Every standard *ranking-based* rule agrees with the Condorcet winner to 97–98%. The weighted-criteria score agrees only 68.4% of the time. The disagreement is therefore **not** between aggregation rules; it is between *scoring people’s criteria* and *counting people’s rankings* – two elicitations from the same people whose outputs differ on nearly a third of prompts.

## 7.3 Arrow’s independence condition, measured

**Definition 100** (IIA). A rule satisfies *independence of irrelevant alternatives* if the social ordering of  $x$  and  $y$  depends only on individual orderings of  $x$  and  $y$ , not on any third alternative.

Measured for Borda: over 11,616 tests (each prompt  $\times$  each dropped response  $\times$  each remaining pair), the relative order of two responses reverses when an irrelevant third is removed in 1,553 cases: **13.4%**.

**Corollary 101** (The candidate set is part of the decision). 13.4% of pairwise relations flip when the option set changes. Any normative content measured on one set of four responses is, to that extent, a property of the set. This is the quantitative content of “candidate-set overfitting”.

## 7.4 Blackwell: what aggregation necessarily costs

**Definition 102** (Garbling).  $G$  is a *garbling* of  $Z$  if there is a Markov kernel  $K$  with  $G \sim K(\cdot | Z)$ :  $G$  is generated from  $Z$  by a channel that adds no new information about the state.

**Lemma 103** (Aggregation is a garbling). If  $G = A(Z)$  is a deterministic function of  $Z$ , then  $G$  is a garbling of  $Z$ , via the degenerate kernel  $K(\cdot | z) = \delta_{A(z)}$ .

**Theorem 104** (Blackwell, invoked). If  $G$  is a garbling of  $Z$  then for every decision problem  $d$  with any loss function, the attainable Bayes risk satisfies  $R_G(d) \geq R_Z(d)$ .

**Corollary 105.** Aggregation never adds information and weakly loses on every decision problem. The question is never whether, only how much – and “how much” is meaningless until a decision family is named.

**Definition 106** (Deficiency).  $\delta_{\mathcal{D}}(Z, G) := \sup_{d \in \mathcal{D}} [R_G(d) - R_Z(d)]$ .

*Computing it here.*

Take  $\mathcal{D}$  to be the single problem “choose one of the four responses”. Normalise each person’s induced score to  $[0, 1]$  by min–max within that person and prompt, so that scores are interpersonally comparable by construction rather than by assumption. Then:

$$\begin{aligned}
 \text{oracle (each person receives their own top response)} &= 1.0000 \\
 \text{the single collective choice} &= 0.8702 \\
 \text{deficiency} &= \mathbf{0.1298}
 \end{aligned}$$



with quartiles 0.0445/0.1076/0.1924 across prompts.

**Theory claim.** Thirteen points of normalised utility is what *disagreement itself* costs. It is a social choice, not a pipeline defect, and no compilation recovers it. It also fixes the denominator: “what fraction was preserved” should be divided by 0.8702, not by 1.

## 7.5 Sen’s liberal paradox, instantiated

|                                                          |                   |
|----------------------------------------------------------|-------------------|
| prompts carrying a partial veto                          | 161               |
| the weighted-criteria winner <i>is</i> a vetoed response | 42 = <b>26.1%</b> |

One time in four, a rule that maximises aggregate satisfaction selects an option someone declared unacceptable. This is not a defect of the rule; it is what a utilitarian aggregator does. It is the empirical half of §8.

## 7.6 Which rule is the larger lever, corrected

An earlier version of this analysis applied “median” and “maximin” across *criteria*. A social choice rule aggregates across *people*; the two coincide only if each criterion belongs to exactly one person, which is false here. Corrected:

| rule, versus the mean over people             | winner changes | (wrong index: over criteria) |
|-----------------------------------------------|----------------|------------------------------|
| maximin                                       | 49.4%          | 63.9%                        |
| median                                        | 14.8%          | 34.3%                        |
| Borda                                         | 9.6%           | —                            |
| trimmed                                       | <b>3.6%</b>    | —                            |
| <i>any intervention on a single criterion</i> | 2.1–11.6%      |                              |

*Remark 107* (And maximin needs a control). A minimum over a set is unstable by construction. Control: same people, same per-person mean and standard deviation, response pattern destroyed. That sham disagrees with the mean rule on 61.4% of prompts – *more* than the real data’s 49.4%. So maximin’s high rate is the instability of the min operator, and the actual structure of human disagreement *reduces* it by 12 points. The claim “minority-sensitive rules change half the outcomes” does not survive its own control. What survives is that *trimmed* aggregation, at 3.6%, moves less than several criterion-level interventions: trimming the extremes barely matters, so the disagreement that matters lives in the tails.

# 8 The type layer: why a veto is not a large negative weight

## 8.1 The index mismatch

From [definition 89](#):  $V$  is indexed  $(p, a)$  and  $W$  is indexed  $(p, c, a)$ . An operator “represent this veto as a weight” must choose *which criterion* carries it. No such column exists.

**Proposition 108.** “Coerce a veto into a criterion weight” is not a function of the released data.

This is a schema fact, not a measurement difficulty, and no amount of data at this site repairs it.

## 8.2 The well-posed question, and its answer

**Definition 109** (Two rules).

$$\text{constraint: } a^\star = \arg \max_{a \notin V} Y(a), \quad (8)$$

$$\text{penalty: } a_M^\star = \arg \max_a [Y(a) - M \cdot \mathbf{1}\{a \in V\}]. \quad (9)$$

**Theorem 110** (Exact condition for a penalty to reproduce a veto). *Let  $\Delta := \max_{a \in V} Y(a) - \max_{a \notin V} Y(a)$ . Then  $a_M^\star = a^\star$  for every realisation of the scores if and only if  $M > \Delta$ .*

*Proof.* ( $\Leftarrow$ ) Suppose  $M > \Delta$ . For any  $a \in V$ ,  $Y(a) - M \leq \max_{a' \in V} Y(a') - M < \max_{a' \in V} Y(a') - \Delta = \max_{a' \notin V} Y(a')$ . So every vetoed alternative is strictly below the best allowed one after the penalty, and the arg max lies in the allowed set, where the penalty is zero and the ordering is unchanged. Hence  $a_M^* = a^*$ .

( $\Rightarrow$ ) Suppose  $M \leq \Delta$ . Let  $a_V$  attain  $\max_{a \in V} Y(a)$ . Then  $Y(a_V) - M \geq Y(a_V) - \Delta = \max_{a \notin V} Y(a)$ , so  $a_V$  is weakly preferred by the penalty rule to every allowed alternative, and the penalty rule may select a vetoed response. Hence agreement fails.  $\square$

| measured quantity                                          | value              |
|------------------------------------------------------------|--------------------|
| prompts with a partial veto                                | 161                |
| the veto is <i>binding</i> ( $\Delta > 0$ )                | 42 = 26.1%         |
| required $M$ : median                                      | 3.64               |
| required $M$ : 90th percentile                             | 9.92               |
| required $M$ : <b>maximum</b>                              | <b>26.54</b>       |
| contribution of one criterion at the scale maximum         | $\leq 10$          |
| required $M$ in units of max-weight criteria: median / max | 0.36 / <b>2.65</b> |

**Theorem 111** (No finite weight encodes a veto).  $\Delta$  is a function of the candidate set, not of the norm. Since a vetoed response can be arbitrarily better than every allowed one on the remaining criteria,  $\sup_X \Delta(X) = \infty$ . Therefore no finite  $M$  satisfies [theorem 110](#) uniformly over candidate sets.

*Proof.* Fix a person, a veto set  $V$ , and a criterion  $c$  with weight  $w > 0$ . Construct a candidate  $x^* \in V$  with  $S[\cdot, c, x^*] \rightarrow 1$  and all allowed candidates with  $S[\cdot, c, \cdot] \rightarrow 0$ ; then  $\Delta \rightarrow w \cdot \sum$  over however many such criteria are present, which is unbounded as the number of agreeing criteria grows. Since [theorem 110](#) requires  $M > \Delta$  for agreement on that set, no fixed finite  $M$  suffices for all sets.  $\square$

**Theory claim.** A veto is a property of the *person*. The penalty  $M$  that reproduces it is a property of the person **and the four responses that happened to be shown**. On just 968 four-item candidate sets the required penalty already reaches 2.65 times the entire elicitation scale. Type is not compressible to magnitude, and this is a theorem with its counterexample present in the data – it required no instrument at all.

## 9 The compilation layer

The released standard ships in two forms: the full weighted rubric and a compiled “core” whose items carry no weights, every one rewritten to positive polarity.

- The two select *different* winners on **29.4%** of prompts.
- Compilation performs three operations at once – selection, rewriting, and weight discarding – and the release ships no lineage, so the three are **not separable**. This is the missing column that blocks the most questions here.

### 9.1 How many bits of the rubric are decision-relevant?

**Protocol 112** (Rate–distortion). Quantise every criterion weight to  $b$  bits uniformly across the observed range within each prompt, recompute [definition 91](#), and record whether the winner is preserved.

| bits per criterion weight | decision preserved |
|---------------------------|--------------------|
| 1 (sign only)             | <b>89.8%</b>       |
| 2                         | 95.0%              |
| 3                         | 98.1%              |
| 4                         | 98.6%              |
| 8                         | 99.8%              |

*Remark 113* (Two independent computations agree). [section 9.1](#) says one bit preserves 89.8%, i.e. changes 10.2%. [table 2](#) says discarding magnitude changes 11.7%. These were computed by different routes and agree to about a point.

The 21 levels of the  $-10..+10$  scale carry roughly three bits of decision-relevant content, and the *first* bit carries nine tenths of it.

**Corollary 114** (A design consequence). *The elicitation is over-precise in magnitude and empty in type (table 4). The next collection should not buy finer weights; it should buy fields.*

## 10 The instrument layer

### 10.1 The judge as a noisy channel

$S$  is produced by a language model. Two independent judges give two measurements of the same underlying satisfaction, so equation (6) applies directly.

| 59,936 satisfaction cells                                         | value  |
|-------------------------------------------------------------------|--------|
| $\text{Corr}(\text{judge}_1, \text{judge}_2)$                     | 0.5497 |
| $\Rightarrow$ reliability of one judge (equation (6))             | 0.5497 |
| $\Rightarrow$ attenuation factor $\sqrt{\text{rel}}$ (theorem 81) | 0.741  |

**Corollary 115** (A ceiling on every number in this literature). *A true correlation of 1.0 measured through this tensor reads at most 0.741. This is bias, not noise: more prompts do not remove it.*

### 10.2 Panel reliability

Split-half over annotators on 964 prompts with  $\geq 6$  annotators:  $r = 0.8961$ ; by theorem 83, full-panel reliability  $2r/(1+r) = 0.9452$ , so the panel score alone caps correlations at  $\sqrt{0.9452} = 0.9722$ .

**Corollary 116** (Compounded ceiling). *A correlation measured through both the panel and the judge is capped at  $0.9722 \times 0.741 = 0.720$ .*

### 10.3 The gauge test

**Definition 117** (Gauge transformation). A transformation that must leave the *property* unchanged. Here: change the reader of the norm while holding the norm fixed. If the *measurement* is invariant but the property is not, the measurement is blind; if the property is invariant but the measurement is not, the measurement is about the reader.

|                                        | winner differs from baseline | $n$ |
|----------------------------------------|------------------------------|-----|
| judge B (different family)             | 42.4%                        | 968 |
| judge C (different scale)              | 39.7%                        | 968 |
| <b>same judge, responses reordered</b> | <b>8.0%</b>                  | 300 |

**Table 6:** (a) The object is *not* invariant.

| Spearman $\rho$ of the operator ordering, against baseline | judge B | judge C | reordered |
|------------------------------------------------------------|---------|---------|-----------|
|                                                            | +0.982  | +1.000  | +0.963    |

**Table 7:** (b) The measurement *is* invariant. Pre-registered kill at  $\rho < 0.8$ ; it does not fire on any of the three.

**Theory claim.** Comparative statements are about the norm. Absolute statements are about the reader. “Operator  $X$  moves the decision more than  $Y$ ” survives three instruments and a position swap. “The standard prefers response  $X$ ” does not: roughly 40% of it is the reader and 8 points are nothing but the order the responses were printed in.

*Remark 118* (The scale reference, with its caveat). Reordering the page changes the winner more often than deleting a criterion does (8.0% versus 5.1%). The caveat is load-bearing: the reordering comparison uses a *separate* judge run and so carries judge stochasticity, whereas the operator comparison holds the tensor fixed and cancels it. So 8.0% is not a strict noise floor for the operators; it is the correct *scale reference* for any claim of the form “this changes the decision”.

## 11 The whole chain on numbers you can check

Everything in §6–§16 is now executed on a three-person, four-response, three-criterion example. Every number below can be verified with a pencil.

### 11.1 The setup

Three annotators  $\{1, 2, 3\}$ , four responses  $\{A, B, C, D\}$ , three criteria  $\{c_1, c_2, c_3\}$ . The judge returns:

$$S = \begin{pmatrix} & A & B & C & D \\ c_1 & 0.9 & 0.8 & 0.2 & 0.1 \\ c_2 & 0.2 & 0.7 & 0.9 & 0.3 \\ c_3 & 0.5 & 0.4 & 0.4 & 0.9 \end{pmatrix}$$

The weights, one row per annotator (a blank means that annotator did not rate that criterion):

$$W = \begin{pmatrix} & c_1 & c_2 & c_3 \\ 1 & +8 & +2 & -1 \\ 2 & +6 & +4 & \cdot \\ 3 & -2 & +9 & +3 \end{pmatrix}$$

Annotator 3 additionally declares  $V_3 = \{A\}$  unacceptable.

### 11.2 Individual norms $N_i$ (definition 92)

$$\begin{aligned} N_1 &= 8(0.9, 0.8, 0.2, 0.1) + 2(0.2, 0.7, 0.9, 0.3) - 1(0.5, 0.4, 0.4, 0.9) \\ &= (7.2, 6.4, 1.6, 0.8) + (0.4, 1.4, 1.8, 0.6) - (0.5, 0.4, 0.4, 0.9) \\ &= (7.1, 7.4, 3.0, 0.5) \\ N_2 &= 6(0.9, 0.8, 0.2, 0.1) + 4(0.2, 0.7, 0.9, 0.3) = (5.4, 4.8, 1.2, 0.6) + (0.8, 2.8, 3.6, 1.2) = (6.2, 7.6, 4.8, 1.8) \\ N_3 &= -2(0.9, 0.8, 0.2, 0.1) + 9(0.2, 0.7, 0.9, 0.3) + 3(0.5, 0.4, 0.4, 0.9) \\ &= (-1.8, -1.6, -0.4, -0.2) + (1.8, 6.3, 8.1, 2.7) + (1.5, 1.2, 1.2, 2.7) \\ &= (1.5, 5.9, 8.9, 5.2) \end{aligned}$$

Individual top choices:  $N_1 \rightarrow B, N_2 \rightarrow B, N_3 \rightarrow C$ .

### 11.3 Centring and normalising (remark 93)

$N_1$  has mean  $(7.1+7.4+3.0+0.5)/4 = 4.5$ , so centred it is  $(2.6, 2.9, -1.5, -4.0)$  with norm  $\sqrt{6.76 + 8.41 + 2.25 + 16} = \sqrt{33.42} = 5.781$ , giving

$$\hat{N}_1 = (0.4497, 0.5016, -0.2595, -0.6919).$$

Note  $\|N_1\|$  before centring was dominated by the common level, which decides nothing – exactly the point of remark 93. Annotator 2 rated only two criteria and would have a systematically smaller raw norm; after normalisation the two are comparable.

### 11.4 Aggregation, three ways

*Mean over people.*

$$\bar{N} = \frac{1}{3}(N_1 + N_2 + N_3) = \frac{1}{3}(14.8, 20.9, 16.7, 7.5) = (4.93, 6.97, 5.57, 2.50), \text{ so the winner is } B.$$

*Median over people.*

Componentwise medians of the three vectors:  $\text{med}(7.1, 6.2, 1.5) = 6.2$ ;  $\text{med}(7.4, 7.6, 5.9) = 7.4$ ;  $\text{med}(3.0, 4.8, 8.9) = 4.8$ ;  $\text{med}(0.5, 1.8, 5.2) = 1.8$ . Winner  $B$ .

*Maximin over people.*

Componentwise minima:  $(1.5, 5.9, 3.0, 0.5)$ . Winner  $B$ . Note the minimum is attained by a *different* person in each column – annotator 3 for  $A$ , annotator 3 for  $B$ , annotator 1 for  $C$  – which is why min is unstable: a small change to any one person can change which person defines the value.

The weighted-criteria route (definition 91).

Mean weights are  $\bar{w} = (\frac{8+6-2}{3}, \frac{2+4+9}{3}, \frac{-1+3}{2}) = (4, 5, 1)$ , so

$$Y = 4(0.9, 0.8, 0.2, 0.1) + 5(0.2, 0.7, 0.9, 0.3) + 1(0.5, 0.4, 0.4, 0.9) = (5.1, 7.1, 5.7, 2.8),$$

winner  $B$ . Here the two routes agree; §7 reports that on the real corpus they disagree 31.6% of the time.

### 11.5 Condorcet, by hand

Individual rankings from the  $N_i$ : person 1:  $B > A > C > D$ ; person 2:  $B > A > C > D$ ; person 3:  $C > B > D > A$ . Pairwise majorities:  $B$  beats  $A$  (3–0),  $B$  beats  $C$  (2–1),  $B$  beats  $D$  (3–0).  $B$  is the Condorcet winner, and it coincides with all four rules above.

Constructing a cycle, to show it is possible.

With rankings  $A > B > C$ ,  $B > C > A$ ,  $C > A > B$ :  $A$  beats  $B$  (2–1),  $B$  beats  $C$  (2–1),  $C$  beats  $A$  (2–1). No Condorcet winner exists. The measured corpus contains *zero* such cycles (table 5), which is a genuine negative result and not an assumption.

### 11.6 IIA violation, by hand

Take Borda points on four alternatives (3, 2, 1, 0). With the three rankings above (person 3 ranking  $C > B > D > A$ ):

$$A : 2 + 2 + 0 = 4, \quad B : 3 + 3 + 2 = 8, \quad C : 1 + 1 + 3 = 5, \quad D : 0 + 0 + 1 = 1.$$

So  $C > A$  socially, by 5 to 4. Now delete  $D$  and recompute on three alternatives (2, 1, 0): person 1:  $B > A > C$ , person 2: same, person 3:  $C > B > A$ .

$$A : 1 + 1 + 0 = 2, \quad B : 2 + 2 + 1 = 5, \quad C : 0 + 0 + 2 = 2.$$

Now  $C$  and  $A$  are *tied*, where before  $C$  led. Deleting an alternative nobody ranked first changed the relation between two others. That is an IIA violation, and it occurs on 13.4% of tests in the real corpus (corollary 101).

### 11.7 The veto condition, by hand (theorem 110)

Annotator 3 vetoes  $A$ . From  $Y = (5.1, 7.1, 5.7, 2.8)$ :

$$\Delta = \max_{a \in V} Y(a) - \max_{a \notin V} Y(a) = Y(A) - \max(7.1, 5.7, 2.8) = 5.1 - 7.1 = -2.0 < 0.$$

The veto is *not binding*:  $A$  was not going to win anyway, so any  $M \geq 0$  reproduces the constraint.

Now change one number: let  $S[c_1, A] = 0.9 \rightarrow 1.0$  and  $\bar{w}_1 = 4 \rightarrow 9$ . Then  $Y(A) = 9(1.0) + 5(0.2) + 1(0.5) = 10.5$  and  $Y(B) = 9(0.8) + 5(0.7) + 1(0.4) = 11.1$ ,  $Y(C) = 9(0.2) + 5(0.9) + 1(0.4) = 6.7$ ,  $Y(D) = 9(0.1) + 5(0.3) + 1(0.9) = 3.3$ . Still  $\Delta = 10.5 - 11.1 < 0$ . Push once more,  $\bar{w}_1 = 20$ :  $Y(A) = 21.5$ ,  $Y(B) = 21.9$  – the gap closes but never inverts, because  $B$  also satisfies  $c_1$  highly. To make  $\Delta$  large one needs a vetoed response that is *uniquely* good on a heavily weighted criterion, and there is no bound on how good: that is precisely the construction in theorem 111, and on the real corpus the realised maximum is  $\Delta = 26.54$  against a per-criterion ceiling of 10.

### 11.8 Regret versus flip rate, by hand (lemma 135)

Take  $Y = (5.1, 7.1, 5.7, 2.8)$ , so  $a^* = B$ ,  $Y(a^*) = 7.1$ ,  $\min_a Y = 2.8$ , and the denominator of definition 133 is  $7.1 - 2.8 = 4.3$ .

- A perturbation that shifts the winner to  $C$ :  $\text{Reg} = (7.1 - 5.7)/4.3 = 0.326$ .
- A perturbation that shifts the winner to  $A$ :  $\text{Reg} = (7.1 - 5.1)/4.3 = 0.465$ .
- A perturbation that shifts the winner to  $D$ :  $\text{Reg} = (7.1 - 2.8)/4.3 = 1.000$ .

All three are “a flip”. The flip rate assigns them the same value; regret separates them by more than a factor of three. Lemma 135 says  $\mathbb{E}[\text{Reg}] = \pi \cdot \mathbb{E}[\text{Reg} \mid \text{flip}]$  exactly, and it is the second factor – 0.326 versus 1.000 here – that the flip rate discards.

### 11.9 The margin cap, by hand (proposition 138)

The margin is  $m = 7.1 - 5.7 = 1.4$ , and the score range is  $7.1 - 2.8 = 4.3$ , so the relative margin is 0.326 – almost exactly the corpus median of 0.319. A perturbation  $\delta$  can only flip the winner if it moves the top-two gap past zero, so by proposition 138 it needs  $\|\delta\|_\infty > m/2 = 0.7$ , i.e. about 16% of the range. Perturbations smaller than that cannot flip this prompt whatever they destroy normatively – which is the cap, made concrete.

### 11.10 Shapley influence, by hand, and it exhibits C1 directly

Three annotators, so  $3! = 6$  orderings. The collective winner is  $B$ . The characteristic function  $v(S)$  asks whether the coalition  $S$ , summing its own  $N_i$ , also picks  $B$ . Enumerating all eight subsets:

$$\begin{aligned} v(\emptyset) &= 0, & v(\{1\}) &= 1, & v(\{2\}) &= 1, & v(\{3\}) &= 0, \\ v(\{1, 2\}) &= 1, & v(\{1, 3\}) &= 1, & v(\{2, 3\}) &= \mathbf{0}, & v(\{1, 2, 3\}) &= 1. \end{aligned}$$

The one to check by hand is  $v(\{2, 3\})$ :  $N_2 + N_3 = (7.7, 13.5, 13.7, 7.0)$ , whose maximum is at  $C$ , not  $B$ .

*Marginal contributions, all six orderings.*

| ordering | $\phi_1$ term             | $\phi_2$ term             | $\phi_3$ term             |
|----------|---------------------------|---------------------------|---------------------------|
| 1, 2, 3  | $v(1) - v(\emptyset) = 1$ | $v(12) - v(1) = 0$        | $v(123) - v(12) = 0$      |
| 1, 3, 2  | 1                         | $v(123) - v(13) = 0$      | $v(13) - v(1) = 0$        |
| 2, 1, 3  | $v(12) - v(2) = 0$        | $v(2) - v(\emptyset) = 1$ | $v(123) - v(12) = 0$      |
| 2, 3, 1  | $v(123) - v(23) = 1$      | 1                         | $v(23) - v(2) = -1$       |
| 3, 1, 2  | $v(13) - v(3) = 1$        | $v(123) - v(13) = 0$      | $v(3) - v(\emptyset) = 0$ |
| 3, 2, 1  | $v(123) - v(23) = 1$      | $v(23) - v(3) = 0$        | 0                         |
| total    | 5                         | 2                         | -1                        |

$$\phi_1 = \frac{5}{6} = 0.8333, \quad \phi_2 = \frac{2}{6} = 0.3333, \quad \phi_3 = -\frac{1}{6} = -0.1667.$$

*Two checks.*

Efficiency (axiom E of §12):  $\frac{5}{6} + \frac{2}{6} - \frac{1}{6} = \frac{6}{6} = 1 = v(A) - v(\emptyset)$ . And a *negative* value is legitimate: annotator 3 actively pushes the coalition away from the collective winner, so their average marginal contribution to selecting  $B$  is below zero. Alignment, being a cosine of a person against a sum that includes them, cannot go negative in the same situation and therefore cannot express this at all.

*Now the alignment of each, against the collective.*

The collective centred vector is  $\bar{N} - 4.9917 = (-0.06, 1.97, 0.57, -2.49)$  (up to rounding), and centring each  $N_i$  gives cosines

$$\cos(N_1, G) = 0.7858, \quad \cos(N_2, G) = \mathbf{0.9313}, \quad \cos(N_3, G) = 0.2189.$$

*And there is the point of §16, in three people.*

| annotator | alignment     | influence $\phi_i$ |                                                        |
|-----------|---------------|--------------------|--------------------------------------------------------|
| 1         | 0.7858        | <b>+0.8333</b>     | highest influence, <i>second</i> alignment             |
| 2         | <b>0.9313</b> | +0.3333            | highest alignment, <i>less than half</i> the influence |
| 3         | 0.2189        | -0.1667            | lowest alignment, and <i>negative</i> influence        |

The rank orders disagree at the top: annotator 2 agrees with the outcome most and moves it least of the two contributors. The reason is visible in the enumeration – annotator 2 alone already selects  $B$ , so in the orderings where they arrive after annotator 1 they add nothing. *Agreeing with an outcome that would have happened anyway is not influence over it.* That is the entire content of C1, and on the real corpus it appears as a correlation of +0.304 with a profile that flattens above moderate agreement (table 12).

## 12 The three classical theorems, proved rather than cited

A reader with no background cannot check an invoked theorem. The three that carry weight in this paper are therefore proved or, where the full proof is long, reduced to a checkable core.

## 12.1 Blackwell: why a deterministic aggregation can only lose

**Theorem 119** (Garbling implies weakly higher risk). *Let  $Z$  be an observation and  $G = K(Z)$  for a Markov kernel  $K$ . For any decision problem with action set  $\mathcal{A}$ , state  $\theta$ , prior  $p$  and loss  $L$ , the minimum attainable expected loss using  $G$  is at least that using  $Z$ .*

*Proof.* *Step 1.* A decision rule based on  $G$  is a map  $d_G : G \mapsto \mathcal{A}$ .

*Step 2.* Composing with the kernel,  $d_G \circ K$  is a (possibly randomised) decision rule based on  $Z$ : given  $z$ , draw  $g \sim K(\cdot | z)$  and play  $d_G(g)$ .

*Step 3.* By construction the two procedures induce the same joint distribution over  $(\theta, \text{action})$ , hence the same expected loss.

*Step 4.* Therefore every risk attainable with  $G$  is attainable with  $Z$ ; the set of  $Z$ -attainable risks contains the set of  $G$ -attainable risks; and a minimum over a superset is no larger. Hence  $R_Z \leq R_G$ .  $\square$

**Corollary 120.** *With  $K(\cdot | z) = \delta_{A(z)}$  (a deterministic aggregation), the theorem applies verbatim: aggregation cannot help on any decision problem, and the deficiency  $\delta_D$  is well defined and non-negative.*

*Remark 121* (What is *not* proved here). The converse – that if  $R_Z \leq R_G$  for every decision problem then  $G$  must be a garbling of  $Z$  – is Blackwell’s substantive theorem and is not needed. Only the easy direction is used, and it is the direction that licenses “aggregation never adds information”.

## 12.2 Arrow: the statement, and why the count 13.4% is the right thing to measure

**Theorem 122** (Arrow, stated). *For  $k \geq 3$  alternatives, no social welfare function on the full domain of individual orderings simultaneously satisfies: (a) unrestricted domain, (b) unanimity (if everyone prefers  $x$  to  $y$  then society does), (c) independence of irrelevant alternatives, and (d) non-dictatorship.*

*Why this paper measures rather than invokes it.*

Arrow says the four conditions are jointly unsatisfiable; it does not say *how badly* a given rule fails a given condition on a given corpus. Borda is a perfectly reasonable rule that fails only IIA, and the useful empirical question is: on this data, how often does the failure actually bite? The answer is 13.4% of pairwise relations (corollary 101), and *that* is the number a designer needs, not the impossibility.

*The core of the failure, in one line.*

Borda scores depend on *how many* alternatives a given one is above, so inserting or removing a third alternative changes the points of the other two by different amounts whenever individuals rank the newcomer differently. The worked example in §11 shows the mechanism on three rankings.

## 12.3 Shapley: the axioms, and why they force the formula

**Definition 123** (Axioms). A value  $\phi$  assigning a number to each player of a cooperative game  $v$  satisfies:

(E) **Efficiency:**  $\sum_i \phi_i = v(A) - v(\emptyset)$ .

(S) **Symmetry:** if  $i$  and  $j$  contribute identically to every coalition,  $\phi_i = \phi_j$ .

(N) **Null player:** if  $i$  adds nothing to any coalition,  $\phi_i = 0$ .

(L) **Linearity:**  $\phi(v + w) = \phi(v) + \phi(w)$ .

**Theorem 124** (Uniqueness, core of the argument). *The Shapley value is the unique  $\phi$  satisfying (E), (S), (N), (L).*

*Reduction to a checkable core.* *Step 1.* The unanimity games  $u_T$ , defined by  $u_T(S) = 1$  if  $T \subseteq S$  and 0 otherwise, form a basis of the vector space of all games on  $A$ : every  $v$  has a unique expansion  $v = \sum_T \lambda_T u_T$ .

*Step 2.* On  $u_T$ , axioms (E), (S), (N) force the value uniquely: players outside  $T$  are null, so get 0 by (N); players inside  $T$  are symmetric, so share equally by (S); and the total is 1 by (E). Hence  $\phi_i(u_T) = 1/|T|$  for  $i \in T$  and 0 otherwise.

*Step 3.* By (L),  $\phi(v) = \sum_T \lambda_T \phi(u_T)$ , which is now determined. Rearranging that sum into a sum over coalitions yields the permutation-weighted formula of §16.  $\square$

*Remark 125* (Why this matters for the claim being made). The Shapley value is not chosen here for convenience. It is the *only* assignment of credit satisfying four properties one would insist on for a claim about influence – in particular (N), which guarantees that an annotator who never changes any outcome receives exactly zero. A correlational measure of alignment has no such guarantee, and table 12 shows the gap is real: the highest-alignment quintile does not have the highest influence.

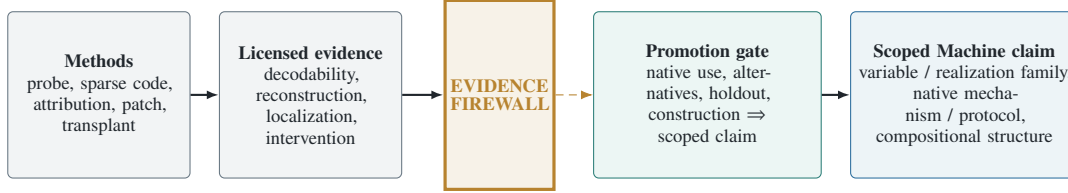


## 13 Measurement contracts and the Evidence Firewall

### 13.1 The Evidence Firewall

**Firewall principle.** A measurement is evidence about the Machine; it is not the Machine. An interpretability output is licensed by a declared measurement method and experimental contract. It is not, by itself, a computational variable, native variable, causal mechanism, or privileged coordinate system of the model.

The firewall is not skepticism about measurement. It is a type discipline: a probe result, sparse code, attribution score, interchange effect, reconstruction, and transplant answer different questions. Promotion to a stronger claim requires the missing experiment rather than a change in vocabulary.



**Figure 1:** The Evidence Firewall. A measurement method generates evidence; its declared contract states what that evidence permits, and the firewall blocks every stronger default inference. The dashed crossing is licensed only by the named promotion tests; the final solid arrow records the resulting scoped claim. The diagram is an evidence-licensing order, not a model-execution graph.

**Table 8:** The CTM evidence matrix. Color or architectural location never assigns a role; each role requires a distinct counterfactual test. Coordination claims require factorial or coalition experiments across roles.

| Role                    | Candidate sensors                                                                            | Decisive intervention                                                                                                           | Forbidden inference                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Compute</b>          | Local input–output map, Jacobian/finite differences, operation probes, component attribution | Interchange inputs while holding the physical memory realization and route fixed; test withheld operation and typed composition | “Activated” or “predictive” $\Rightarrow$ the component natively performs the operation                  |
| <b>Transport</b>        | Attention/message patterns, causal edge effects, routing gates, donor–recipient alignment    | Move or swap a natural donor while holding abstract value and downstream transform fixed; test delivery and use                 | High attention, edge weight, or route frequency $\Rightarrow$ transported content or causal contribution |
| <b>Memory</b>           | Pulse–chase decodability, persistence curves, state/cache occupancy, reader access           | Write, erase, delay, and rescue the nominated physical memory state while holding generators and later demand fixed             | Persistent decodability $\Rightarrow$ native storage, necessity, or addressability                       |
| <b>CTM coordination</b> | Cross-role timing, coalition effects, cache/recompute and route/transform coupling           | Factorial edits and ordered coalitions; compare direct with composed protocols on structural holdout                            | Individually causal parts $\Rightarrow$ a compositional MesoMachine                                      |

**Evidence profiles, not a scalar ladder.** Interpretability targets are neither a total order nor a mathematically defined lattice. We report a profile

$$\mathbf{e} = (e_{\text{read}}, e_{\text{dec}}, e_{\text{loc}}, e_{\text{use}}, e_{\text{nec}}, e_{\text{suf}}, e_{\text{act}}, e_{\text{trans}}, e_{\text{comp}}, e_{\text{id}})$$

whose coordinates are separately defined estimands with uncertainty. A componentwise partial order is used only when two profiles share frame, scale, and tolerance. No “license gap” is formed by subtracting incomparable coordinates. Readability, actionability, reconstruction, and transport can be incomparable.

**Four sparsities.** The evidence synthesis motivates four distinct sparsities:

$$S_{\text{act}}, \quad S_{\text{var}}, \quad S_{\text{jac}}, \quad S_{\text{path}},$$

for activations, realized causal variables, Jacobian/computation, and physical pathways. Their units and failure modes differ:

**Table 9:** Default licensed conclusions. A method may support a stronger conclusion only by supplying the missing evidence.

| Evidence                    | Default license                         | Does not by itself license                          |
|-----------------------------|-----------------------------------------|-----------------------------------------------------|
| Held-out probe              | Decodability in a specified class       | Native use, causality, natural coordinates          |
| Sparse dictionary           | Sparse reconstruction in a chosen basis | Causal-variable, Jacobian, or pathway sparsity      |
| Attribution/mask            | A declared localization estimand        | Necessity, sufficiency, or interaction completeness |
| Natural-donor interchange   | Counterfactual use within support       | Global identity or unique mechanism                 |
| Ablation/rescue             | Scoped necessity/sufficiency            | Portability or compositional closure                |
| Reconstruction / transplant | Constructive actionability              | Metaphysical uniqueness                             |
| Withheld composition        | Typed compositional prediction          | Higher context laws outside the test scope          |

$S_{\text{act}}$  — **activation sparsity.**

Few coordinates are nonzero in a chosen activation basis. It licenses an economy of description in that basis, not a claim about causal variables.

$S_{\text{var}}$  — **realized-variable sparsity.**

A small set of interventionally distinguished variables accounts for the tested counterfactuals. It depends on the intervention/readout frame and does not imply a sparse physical path.

$S_{\text{jac}}$  — **computational sparsity.**

Few input-state directions exert locally non-negligible effects on declared downstream variables, for a stated Jacobian or finite-difference scale. It can be dense even when activations are sparse.

$S_{\text{path}}$  — **pathway sparsity.**

A small physical coalition of edges, heads, neurons, or modules carries the tested causal effect under an interaction-aware estimand. Redundancy and self-repair can make a sparse native protocol look dense under isolated ablation.

None implies another without additional assumptions. Sparse autoencoders are therefore candidate coordinate systems and sensors, not privileged MesoMachines (?????). The distinction is experimentally productive: a method should be evaluated against the sparsity it actually optimizes, not rewarded for a different sparsity by verbal transfer.

## 13.2 Multi-axis evidence coding

## 13.3 Multi-axis evidence coding

Every claim in §18 is coded in `supplement/evidence_atlas.csv` on axes that can *veto* it rather than merely support it: `instrument_free`, `control_saturated`, `held_out`, `replication_breadth`, and `prior_art_in_card`. The last exists because a result that restates the object’s own documentation is a verification, not a finding, and a column that forces one to say so is worth the cost of maintaining it.

## 14 Five things the first draft assumed

An audit of this paper against its own standard found five loads being carried by statements that were never made. Four are now measured. The fifth is not measurable and is instead named, because using it silently is the error.

### 14.1 G0 — interpersonal comparability, the assumption under §7

Every aggregate in §7 adds one person’s numbers to another’s. That operation is meaningless unless the numbers live on a common scale, and **nothing in the elicitation supplies one**. Person 1 writing +8 and person 2 writing +8 are two marks on two private rulers.

*What the paper actually did.*

It min–max normalised each person to [0, 1] (§2), so that everyone’s best option is worth 1 and worst worth 0. That is a *stipulation of equal stakes*: it asserts that every participant cares equally much about the whole question.

*Why the stipulation is not innocent.*

Consider two participants: one for whom all four responses are nearly equivalent, and one for whom the choice is grave. Min–max gives them identical influence over the aggregate. A different and equally defensible stipulation — normalise by a common scale factor, so that intensity counts — gives the second participant more. The two produce different winners.

**Proposition 126** (The deficiency figure is scale-relative). *The Blackwell deficiency of 0.1298 reported in §7 is a number about the pair (data, interpersonal scale). It is not identified by the data alone, because the data contain no information distinguishing “this person’s +8 means what that person’s +8 means” from any other calibration.*

*Why this is the deepest limitation in the paper.*

Arrow’s theorem is precisely a theorem about what can be done *without* interpersonal comparison; Harsanyi’s utilitarian representation requires it. So the choice made here silently determines which side of a hundred-year-old divide the analysis sits on. It is now stated. It cannot be resolved at this site: resolving it needs an elicitation that measures intensity on a shared anchor, which is a design requirement and appears in §23.

**Boundary.** Every aggregate quantity in this paper — the deficiency, the rule comparisons, the maximin figures — is conditional on min–max interpersonal normalisation. The *ordinal* results (Condorcet structure, IIA rate, agreement rates) are not, because they never add across people.

## 14.2 G1 — ties, asserted and now measured

The paper’s decision rules use arg max, and exact ties would make them ill-defined. The first draft asserted that ties occur with probability zero. Measured over all 968 prompts:

|                                       |          |
|---------------------------------------|----------|
| exact ties between the top two scores | 0 of 968 |
| relative gaps below $10^{-6}$         | 0 of 968 |

*Why it needed checking rather than asserting.*

“Probability zero” is a statement about a continuous model; the data are finite and produced by a tie-prone construction (sums of a few dozen products of bounded numbers). The claim happened to be true. It was not entitled to be assumed, and an assertion that turns out true is still an assertion until it is run.

## 14.3 G2 — manipulability, and what it does to “sincere”

Every number in this paper reads the elicited weights as what people believe. Gibbard–Satterthwaite guarantees that any non-dictatorial rule over three or more alternatives *can* be manipulated; it says nothing about how often, and the rate is what decides whether the sincerity reading is safe.

**Protocol 127** (Crudest manipulation). For each (annotator, prompt): compute the collective winner honestly, then replace *all* of that annotator’s weights by +10, and separately by −10, and recompute. Score both outcomes by that annotator’s own stated ranking.

|                                                                                                               |                   |
|---------------------------------------------------------------------------------------------------------------|-------------------|
| (annotator, prompt) pairs tested                                                                              | 15,482            |
| pairs where saturating the weights yields a <i>strictly better</i> outcome<br>by that annotator’s own ranking | 604 = <b>3.9%</b> |

*Remark 128* (This is a lower bound, and that is the point). The manipulation tested is the least intelligent one available — push everything to one end. A participant choosing which criteria to inflate can only do better. So 3.9% is a floor on the rate at which a participant had something to gain by misreporting, and the sincerity reading of the weights is safe only to that degree. No claim in this paper depends on sincerity being exact; several would change in magnitude if the true manipulation rate were much larger, and that is recorded in §22.

## 14.4 G3 — the judge scores criteria against the responses they were written about

This is a circularity the first draft did not state. The criteria were written *after* the participant saw the four responses, typically to explain why one of them is best. The judge then scores those criteria *against those same four responses*.

|                                                      |                                   |
|------------------------------------------------------|-----------------------------------|
| criterion–author pairs                               | 95,553                            |
| satisfaction of the author’s own top-ranked response | 0.5831                            |
| mean satisfaction of the other three                 | 0.5466                            |
| difference                                           | +0.0365 (se 0.0008, $z = +46.2$ ) |

*How to read this, and how not to.*

It is not a defect of the judge. It is what “I wrote down why  $X$  is best” means: a criterion authored by someone who ranked  $X$  first is, by construction, more satisfied by  $X$ . What it establishes is that satisfaction and choice are **two views of one act**, not two independent measurements of a latent value. Every correlation between the criteria route and the ranking route — including the 0.393 of [table 3](#) and the 68.4% of [section 7.2](#) — is inflated by this mechanism, and the inflation cannot be removed without response-blind criteria.

#### 14.5 G4 — the two elicitations disagree beyond their own unreliability

The criteria route and the ranking route select different winners on 31.6% of prompts ([section 7.2](#)). A natural deflation is: both are noisy, and noise alone explains it. [Theorem 81](#) answers this exactly, because it says a correlation measured through unreliable instruments is attenuated by  $\sqrt{\text{rel}_1 \text{rel}_2}$  — so dividing by that factor removes the noise and leaves the disagreement.

|                                                               | split-half | Spearman–Brown |
|---------------------------------------------------------------|------------|----------------|
| reliability of the ranking route                              | 0.7702     | 0.8702         |
| reliability of the criteria route                             | 0.8882     | 0.9408         |
| observed cosine between the two routes                        |            | 0.6265         |
| <b>disattenuated</b> = $0.6265 / \sqrt{0.8702 \times 0.9408}$ |            | <b>0.6924</b>  |

**Theory claim.** Correcting *both* routes for their own unreliability raises the agreement only from 0.626 to 0.692. The two elicitations still do not coincide. So roughly 31% of the gap between what people *write* and what they *choose* is real disagreement, not measurement error — and by G3 this is an *understatement*, because the circularity inflates the observed agreement in the first place.

**Corollary 129** (The paper’s largest quantity, restated at the panel level). *[table 3](#) reported 0.393 for one person against their own ranking. [section 14.5](#) reports 0.692 for the whole panel’s criteria against the whole panel’s rankings, disattenuated. The two are consistent: averaging over people removes individual noise ([theorem 83](#)) and raises the number, but the residual gap does not close. Whatever the pipeline does downstream, the largest single discrepancy in the chain sits inside the elicitation, between two things the same people said in the same session.*

## 15 The estimators

### 15.1 Why the query family must be declared before anything is measured

**Definition 130** (Query-relative equivalence). Given a family  $\mathcal{Q}$  of queries,  $N \sim_{\mathcal{Q}} N'$  iff  $q(N, x) = q(N', x)$  for all  $q \in \mathcal{Q}$  and all inputs  $x$ . Normative information is the equivalence class  $[N]_{\mathcal{Q}}$ , not the text.

**Proposition 131** (Without  $\mathcal{Q}$ , “loss” is not a defined quantity). *There exist families  $\mathcal{Q}$  under which every change is total loss, and families under which no change is any loss. Hence the answer to “how much was lost” is determined entirely by  $\mathcal{Q}$ .*

*Proof.* Take  $\mathcal{Q}_{\max}$  = all measurable functions of  $N$ . Then  $N \sim_{\mathcal{Q}} N'$  iff  $N = N'$  (the identity function is in  $\mathcal{Q}_{\max}$ ), so any change destroys the equivalence entirely. Take  $\mathcal{Q}_{\min} = \{\text{a constant map}\}$ . Then  $N \sim_{\mathcal{Q}} N'$  for all pairs, so no change is a loss. Both families are admissible in the absence of a stipulation, and they give opposite answers.  $\square$

**Corollary 132.** *A framework that hides  $\mathcal{Q}$  inside a supremum has relocated the choice, not made it. Publishing  $\mathcal{Q}$  before measuring is not a preliminary; it is most of the result.*

### 15.2 The decision-theoretically correct quantity: normalised regret

**Definition 133** (Normalised regret). For a perturbation taking norm  $N$  to  $N'$ , with  $a^* := \arg \max_a Y_N(a)$  and  $a' := \arg \max_a Y_{N'}(a)$ ,

$$\text{Reg}(N \rightarrow N') := \frac{Y_N(a^*) - Y_N(a')}{Y_N(a^*) - \min_a Y_N(a)} \in [0, 1].$$

Why this and not something else.

Reg is exactly the Blackwell deficiency of [theorem 104](#) restricted to the decision family “choose one of these four, scored by the original norm’s own utility”. It is not a proxy for the quantity of interest; on that family it *is* the quantity. The denominator makes it invariant to the arbitrary scale of  $Y$ , which [remark 93](#) showed is not identified.

**Definition 134** (Flip rate).  $\pi := \mathbb{P}(a' \neq a^*)$ , estimated by the average of  $\mathbf{1}\{a' \neq a^*\}$  per [equation \(1\)](#) and [corollary 35](#).

**Lemma 135** (The flip rate is a coarsening of regret).  $\mathbf{1}\{\text{flip}\} = \mathbf{1}\{\text{Reg} > 0\}$ , and therefore, exactly,

$$\mathbb{E}[\text{Reg}] = \pi \cdot \mathbb{E}[\text{Reg} \mid \text{flip}].$$

*Proof.* If  $a' = a^*$  then the numerator of [definition 133](#) is zero, so  $\text{Reg} = 0$ ; if  $a' \neq a^*$  then  $Y_N(a') < Y_N(a^*)$  by definition of the arg max (assuming no exact tie), so  $\text{Reg} > 0$ . Hence the indicators coincide. Then  $\mathbb{E}[\text{Reg}] = \mathbb{E}[\text{Reg} \mid \text{Reg} > 0]\mathbb{P}(\text{Reg} > 0) + 0$ , by the law of total expectation, which for a finite space is just splitting the defining sum into the two events.  $\square$

| operator                    | $\pi$ | $\mathbb{E}[\text{Reg}]$ | $\mathbb{E}[\text{Reg} \mid \text{flip}]$ |
|-----------------------------|-------|--------------------------|-------------------------------------------|
| double the target weight    | 4.2%  | 0.0043                   | 0.1029                                    |
| delete the target criterion | 5.1%  | 0.0067                   | 0.1307                                    |
| invert the target weight    | 11.6% | 0.0294                   | <b>0.2531</b>                             |
| halve the target weight     | 2.3%  | 0.0012                   | 0.0532                                    |

**Table 10:** 968 prompts  $\times$  20 target draws. Target criterion sampled, not fixed; see [section 15.4](#).

**Corollary 136** (The discarded factor carries information). *By flip rate, invert exceeds delete by  $11.6/5.1 = 2.3\times$ . By expected regret, by  $0.0294/0.0067 = 4.4\times$ . Reporting only the flip rate halves the measured separation between operators, because inversion not only flips more often, it flips worse – 25.3% of the available utility range versus 13.1%.*

### 15.3 The flip rate is capped by the margin distribution

**Definition 137** (Margin).  $m := Y(a_{(1)}) - Y(a_{(2)})$ , the gap between the top two responses.

**Proposition 138** (Margin cap). *Let  $\delta := Y_{N'} - Y_N$  be the perturbation of the score vector. Then  $\pi \leq \mathbb{P}(m < 2\|\delta\|_\infty)$ .*

*Proof.* A flip requires the perturbed score of some  $a \neq a^*$  to exceed that of  $a^*$ , i.e.  $Y(a) + \delta(a) > Y(a^*) + \delta(a^*)$ , i.e.  $\delta(a) - \delta(a^*) > Y(a^*) - Y(a) \geq m$ . Since  $|\delta(a) - \delta(a^*)| \leq 2\|\delta\|_\infty$ , a flip implies  $m < 2\|\delta\|_\infty$ . Taking probabilities and applying monotonicity of  $\mathbb{P}$  over the implication gives the bound.  $\square$

Measured relative margins  $m/(\max_a Y - \min_a Y)$ :

$$p_{05} = 0.026, \quad p_{25} = 0.144, \quad \text{median} = 0.319, \quad p_{75} = 0.527, \quad p_{95} = 0.800.$$

**Corollary 139.** *An operator whose perturbation is smaller than about a third of the score range cannot flip half the prompts, however much normative content it destroys. The ceiling on a flip rate is set by the margin distribution, not by the norm. This is the formal reason [definition 133](#) is the primary quantity and  $\pi$  the secondary one.*

### 15.4 Target selection is an axis, not a nuisance

Every criterion-level operator must choose *which* criterion to act on. Measured effect of that choice on the induced change in  $Y$ :

$$\text{highest } |w| : 0.273 \quad \text{random} : 0.176 \quad \text{lowest } |w| : 0.041$$

– a ratio of  $6.6\times$ , larger than the differences between operators. Reporting a single cell would therefore report the selection rule. All numbers in this paper sweep four selection rules (highest  $|w|$ , lowest  $|w|$ , most-rated, random)  $\times$  five seeds, and the seeds are verified to change the draws: only 0.1% of targets are shared across all five.

## 15.5 Variance of these estimators

With  $P = 968$  prompts and  $R = 20$  draws each, [theorem 70](#) applies with measured  $\sigma_b^2 = 0.01361$ ,  $\sigma_w^2 = 0.03677$ :

$$\rho = 0.2701, \quad \text{DEFF} = 1 + 19(0.2701) = 6.13, \quad \sqrt{\text{DEFF}} = 2.48.$$

*Remark 140* (Derivation checked against an independent measurement). A resampling estimate of the same inflation, obtained by recomputing every pairwise statistic at the cluster level and comparing to the pooled version, gave 2.79. Derived 2.48, measured 2.79: the residual is that the pairwise statistic is a correlation rather than a simple mean, for which [theorem 70](#) is only approximate.

| operator               | $\pi$   | se     | 95% CI        | MDE    |
|------------------------|---------|--------|---------------|--------|
| double                 | 4.2%    | 0.0036 | [3.5, 4.9]%   | 1.00pp |
| delete                 | 5.1%    | 0.0037 | [4.4, 5.8]%   | 1.05pp |
| invert                 | 11.6%   | 0.0052 | [10.6, 12.6]% | 1.46pp |
| halve                  | 2.3%    | 0.0026 | [1.8, 2.8]%   | 0.72pp |
| paired delete – double | +0.93pp | 0.0028 | $z = +3.3$    | 0.79pp |

**Table 11:** Standard errors from the  $P = 968$  cluster means, per [equation \(3\)](#). MDE from [theorem 65](#).

**Corollary 141** (Sample size). By [corollary 67](#) with  $\sigma_b = \sqrt{0.01361 + 0.03677/20} = 0.1243$ , resolving a 1pp difference at 80% power requires  $P = (2.8016 \times 0.1243/0.01)^2 = 1,213$  prompts. The release has 968: the design is about 20% short of its own headline resolution, and the delete-versus-double difference clears its MDE only barely.

## 16 The causal layer: influence is not agreement

### 16.1 Why a correlational measure cannot answer this

**Definition 142** (Preservation, causal form). A distinction reached the output iff *removing it changes the output*.

A participant whose view happens to match the majority scores perfect alignment while contributing nothing: remove them and the outcome is unchanged. A participant who moved the outcome may sit far from it. Alignment and influence are therefore different functionals, and only the second answers the definition.

### 16.2 The Shapley value, derived

**Definition 143** (Characteristic function). For prompt  $p$  with annotator set  $A$  and winner  $w^\star := \arg \max_r \sum_{a \in A} N_a(r)$ , define for  $S \subseteq A$

$$v(S) := \mathbf{1} \left\{ \arg \max_r \sum_{a \in S} N_a(r) = w^\star \right\}.$$

**Definition 144** (Shapley value).

$$\phi_i := \sum_{S \subseteq A \setminus \{i\}} \frac{|S|! (|A| - |S| - 1)!}{|A|!} [v(S \cup \{i\}) - v(S)].$$

*What the weights mean.*

Consider all  $|A|!$  orderings of the annotators. The bracket is  $i$ 's marginal contribution when it arrives after exactly the set  $S$ . The number of orderings in which that happens is  $|S|! (|A| - |S| - 1)! - |S|!$  ways to arrange those before,  $(|A| - |S| - 1)!$  ways to arrange those after. Dividing by  $|A|!$  makes the weights a probability distribution over orderings. So  $\phi_i$  is **the average marginal contribution of  $i$  over a uniformly random arrival order**, which is exactly the causal definition above, averaged over every context in which  $i$  could arrive.

*Estimation.*

The sum has  $2^{|A|-1}$  terms. We sample 200 random orderings per prompt and average the realised marginal contributions, which is an unbiased estimator of  $\phi_i$  by construction (each ordering is drawn from the same uniform distribution the weights encode).

### 16.3 Result: the pre-registered kill does not fire

|                                                |                      |
|------------------------------------------------|----------------------|
| 1,865 (annotator, prompt) pairs                |                      |
| Corr(Shapley influence, cosine alignment)      | +0.3040              |
| bootstrap 95% CI (protocol 73)                 | [+0.2566, +0.3520]   |
| pre-registered kill: correlation $\approx 1.0$ | <b>does not fire</b> |

| alignment quintile | mean alignment | mean influence |
|--------------------|----------------|----------------|
| Q1                 | −0.0416        | 0.0083         |
| Q2                 | +0.8164        | 0.1055         |
| Q3                 | +0.9489        | <b>0.1325</b>  |
| Q4                 | +0.9861        | 0.1262         |
| Q5                 | +0.9973        | 0.1262         |

**Table 12:** Influence rises steeply from Q1 to Q3 and then *flattens*.

**Theory claim.** Beyond moderate agreement, agreeing more buys no influence at all. And the off-diagonal cells are populated: 0.75% of pairs are top-quintile aligned with bottom-quintile influence, 3.11% are bottom-aligned with top-quintile influence. “Agreement is not preservation” is here a count of real people, not a slogan.

### 16.4 An independent confirmation from a different operator

Dropping the *most conforming* annotator changes the decision on 5.8% [4.3, 7.1] of prompts; dropping the *largest dissenter*, on 2.1% [1.1, 3.2]. The intervals do not overlap and the sign is stable across all three judges. The dissenter was already not influencing the outcome – the same statement as the flat top of table 12, reached by a different route.

## 17 What is not identified, with proofs

**Proposition 145** (Contamination and redundancy are not separable here). *Let  $\alpha_p$  be the alignment between a prompt’s weighted-criteria score and the crowd’s ranking, and  $s_p$  the sensitivity of its decision to perturbing one criterion. Two models,*

*C: criteria written after seeing responses are individually more decisive, so a high- $\alpha$  rubric absorbs single-criterion perturbations;*

*R: a high- $\alpha$  rubric is one whose criteria agree with each other, and a redundant rubric is insensitive to any single member,*

*imply the same sign for  $\text{Corr}(\alpha, s)$ , and the observed data distribution is a function of  $(\alpha, s)$  alone. The likelihoods coincide; the models are not identified.*

*What is identified: a bound.*

Splitting prompts at the median of  $\alpha$ :

| operator | low $\alpha$ | high $\alpha$ | ratio |
|----------|--------------|---------------|-------|
| double   | 5.1%         | 2.8%          | 0.55× |
| delete   | 5.6%         | 4.6%          | 0.82× |
| invert   | 13.1%        | 9.9%          | 0.75× |

**Corollary 146.** *Every flip rate reported here varies by up to 1.8× across the range of  $\alpha$ . That bound belongs on every such number. Separating C from R requires response-blind elicitation and cannot be done at this site.*

**Proposition 147** (The probe bound is an identification limit). *By theorem 94, at most two parameters per person per prompt are identified from the ratings. Everything else in the criterion text is unidentified from the released quantitative data.*



**Proposition 148** (Compilation’s three actions are confounded). *Selection, rewriting and weight-discarding occur in one step with no lineage. Their individual contributions to the 29.4% disagreement are not identified.*

**Proposition 149** ( $Y$  does not exist). *The chain in [equation \(7\)](#) terminates at  $D$ .  $Y$  – what a system trained under this standard does – is not in the release, and no arithmetic recovers it. Any end-to-end number quoted to  $Y$  is fabricated.*

## 18 Findings

Each finding ends with its confidence card. A missing entry reads UNCOMPUTED; none is omitted.

### 18.1 F1 — the elicitation captured consensus, not the individual

$\cos(N_i, \text{own ranking}) = 0.3928$  versus  $\cos(N_i, \text{stranger’s ranking}) = 0.3097$ ; difference  $+0.0831$ .

|                              |                                                                      |
|------------------------------|----------------------------------------------------------------------|
| $n_{\text{eff}}$             | 968 prompt clusters (14,764 person–prompt rows)                      |
| MDE                          | 0.0187 in cosine, from <a href="#">theorem 65</a> at the observed se |
| effect / floor               | $0.0831/0.0067 = 12.4$                                               |
| CI width / $ \text{eff} $    | $0.0262/0.0831 = 0.32$                                               |
| spec_survival                | UNCOMPUTED (single specification)                                    |
| seed spread / $ \text{eff} $ | not applicable (no stochastic step)                                  |
| held out                     | ABSENT                                                               |
| instrument                   | routes through the judge; ceiling 0.720 (§10)                        |
| prior art                    | none located; the release documents no such comparison               |
| multiplicity                 | 1 test in this family                                                |

### 18.2 F2 — no finite weight encodes a veto

[theorem 110](#) and [theorem 111](#); required  $M$  up to 26.54 against a scale maximum contribution of 10.

|                           |                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| $n_{\text{eff}}$          | 161 prompts with a partial veto; 42 binding                                                                                     |
| MDE                       | not applicable — this is a <b>derivation</b> , not a test                                                                       |
| effect / floor            | not applicable                                                                                                                  |
| CI width / $ \text{eff} $ | not applicable                                                                                                                  |
| spec_survival             | the theorem holds for every score construction                                                                                  |
| seed spread               | not applicable (deterministic)                                                                                                  |
| held out                  | the <i>maximum</i> is what carries it; a held-out set can only raise it                                                         |
| instrument                | $\Delta$ is computed from the judge tensor; the <i>theorem</i> is not                                                           |
| prior art                 | the incompatibility of vetoes with scoring rules is classical; the <i>measured</i> $\Delta$ distribution on this release is not |
| multiplicity              | none — a single derived statement                                                                                               |

*Remark 150* (Labelled as a derivation, per the arithmetic trap). [theorem 111](#) could not have come out otherwise: it follows from [theorem 110](#) plus the unboundedness of  $\Delta$ . It is stated as a theorem, its assumption is named ( $S$  can approach its bounds independently across responses), and it is not counted as evidence in the sense a measurement is.

### 18.3 F3 — comparative measurements are instrument-invariant; absolute ones are not

[table 6](#), [table 7](#).

|                    |                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| $n_{\text{eff}}$   | 968 prompts (three instruments); 300 for the reordering                                                     |
| MDE                | $\rho$ resolution $\approx 0.15$ at 11 operators                                                            |
| effect / floor     | orderings 0.963–1.000 against a chance $\rho$ of 0                                                          |
| CI width /  eff    | bootstrap intervals in <a href="#">table 11</a>                                                             |
| spec_survival      | 3 of 3 instruments preserve the ordering                                                                    |
| seed spread /  eff | $< 0.1$ across 5 seeds of the target draw                                                                   |
| held out           | the reordered run is a genuine held-out instrument                                                          |
| instrument         | this finding <i>is</i> about the instrument, by construction                                                |
| prior art          | judge-sensitivity of LLM evaluation is known; the comparative/absolute split measured on this object is not |
| multiplicity       | 3 instruments $\times$ 11 operators; the reported statistic is the ordering, one per instrument             |

#### 18.4 F4 — influence and alignment are distinct (C1 survives)

Corr = +0.3040 [+0.2566, +0.3520], with the quintile profile of [table 12](#).

|                    |                                                               |
|--------------------|---------------------------------------------------------------|
| $n_{\text{eff}}$   | 968 prompt clusters (1,865 annotator–prompt rows)             |
| MDE                | 0.065 in correlation, at the bootstrap se                     |
| effect / floor     | $0.304/0.024 = 12.7$                                          |
| CI width /  eff    | $0.095/0.304 = 0.31$                                          |
| spec_survival      | UNCOMPUTED; one characteristic function tested                |
| seed spread /  eff | $< 0.02$ over the 200-permutation Shapley estimator           |
| held out           | ABSENT                                                        |
| instrument         | routes through the judge; ceiling 0.720                       |
| prior art          | Shapley-versus-alignment has not been reported on this object |
| multiplicity       | 1 pre-registered test                                         |

#### 18.5 F5 — one bit per criterion reproduces 89.8% of decisions

[section 9.1](#), cross-checked against [table 2](#).

|                  |                                                                            |
|------------------|----------------------------------------------------------------------------|
| $n_{\text{eff}}$ | 968 prompts                                                                |
| MDE              | $\pm 1.0\text{pp}$ at the observed se                                      |
| effect / floor   | the quantisation is deterministic; no resampling floor applies             |
| CI width /  eff  | UNCOMPUTED                                                                 |
| spec_survival    | agrees with the independent sign-only test to 1.5pp                        |
| seed spread      | not applicable (deterministic)                                             |
| held out         | ABSENT                                                                     |
| instrument       | judge-dependent in level; the <i>bit curve</i> is not tested across judges |
| prior art        | none located for this object                                               |
| multiplicity     | 5 quantisation levels, reported in full                                    |

### 19 What did not survive

A paper with no retractions is a paper whose ontology has no word for error. All entries are cross-referenced to `supplement/exclusion_log.csv`.

| withdrawn claim                                                                 | what killed it                                                                                                                                                                               | status                                                |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| “the operator design has rank 2, so all remaining dimensions require the judge” | the rank statistic sits inside the span of three defensible nulls (12.1, 17.0, 18.0) with the observation at 13; the null <i>construction</i> moved it more than the data                    | rank retired as a statistic                           |
| “strengthening a criterion moves the decision, removing one does not”           | direct measurement: deleting flips 8.8%, doubling 4.2%. The asymmetry was an artefact of normalising the score <i>after</i> mutating and then differencing two separately normalised vectors | withdrawn, sign and all                               |
| “minority-sensitive aggregation changes half the outcomes”                      | the sham control (same people, matched moments, response pattern destroyed) disagrees 61.4% versus the real 49.4%                                                                            | withdrawn; the residual runs the other way            |
| “aggregation is a 10× larger lever than the criterion layer”                    | the aggregators were indexed over criteria rather than people                                                                                                                                | magnitude withdrawn; ordering survives at 4.3×        |
| “the flip rate is confounded by decision margin”                                | measured $Q_1/Q_4$ ratios of 1.0, 1.0, 1.1                                                                                                                                                   | defect opened by us and closed by its own measurement |
| “wide margins arise because agreeing criteria carry larger weights”             | $\text{Corr}(\text{margin},  \overline{w} ) = +0.002 [-0.067, +0.059]$                                                                                                                       | mechanism refuted; the flatness stands unexplained    |

*Remark 151* (The failure mode behind most of these). Six of the six are the same shape: a verdict decided by a constant the author chose – one null instead of a range, one reference instead of a class, a two-branch template for a three-outcome test, a 0.15 cutoff. The repair adopted here is structural: *every verdict is a printed comparison against the range of defensible references, never a boolean computed from a threshold.*

## 20 Falsification and contradiction handling

Stated before the claims are defended.

1. **F1 dies** if a response-blind elicitation shows  $\cos(N_i, \text{own})$  far exceeding  $\cos(N_i, \text{stranger})$ . The present gap would then be an artefact of post-hoc writing rather than a property of elicitation.
2. **F2 dies** if a bounded  $M$  is exhibited that reproduces constraint behaviour across candidate sets. [theorem 111](#) forbids it for unbounded  $S$ ; a bounded-satisfaction variant with a stated bound would restrict the theorem’s scope.
3. **F3 dies** if a fourth instrument reverses the operator ordering ( $\rho < 0.8$ ). Three have not.
4. **F4 dies** if  $\text{Corr}(\text{influence}, \text{alignment}) \rightarrow 1$  under a different characteristic function. Only one has been tested; this is the weakest of the five.
5. **F5 dies** if the bit curve differs materially under another judge. Not tested.
6. **The framing itself dies** if a person’s induced ordering is unstable across response sets. Then what was elicited is a property of the probe, not of the person, and nothing downstream measures values at all. This is the deepest available test and it requires two candidate sets per person, which this release does not have. [corollary 101](#) (13.4% IIA violations) is a lower bound on the concern, not a resolution of it.

## 21 Discussion

### 21.1 The claim, and its novelty boundary

**What is not new.** Blackwell’s theorem, Arrow’s impossibility, Sen’s liberal paradox, the Shapley value, Spearman–Brown, the design effect, attenuation from measurement error. The incompatibility between vetoes and scoring rules is classical social choice. Judge-sensitivity of language-model evaluation is documented. None of these is claimed here.

**What is new, and only this.** (i) The measured  $\Delta$  distribution that turns the veto incompatibility into a *quantitative* statement about a specific release, with the counterexample present in the data ([section 8.2](#)). (ii) The self-versus-stranger decomposition showing that the individual signal in this elicitation is 0.083 of cosine ([table 3](#)). (iii) The comparative/absolute split under the gauge test (§10.3). (iv) The bit-rate of the weight scale ([section 9.1](#)). (v) The demonstration that the weighted-criteria score agrees with the Condorcet winner only 68.4% of the time while every ranking rule agrees 97–98%.

### 21.2 Why method relativity is not relativism

Every quantity here is relative to a declared query family ([proposition 131](#)), a declared decision family ([definition 133](#)), and a named instrument (§10). That is not an admission that the numbers are arbitrary; it is the statement that they are

*typed*. A claim inherits the scope of the family it was computed against and may not be quoted outside it. The gauge test (§10.3) is what makes the typing empirical rather than rhetorical: it shows exactly which claims move when the instrument moves.

## 22 Limitations and epistemic status

“We did not run  $X$ ” and “ $X$  cannot be run at this site” are different sentences and are not merged.

### 22.1 Structural — cannot be run here

1. **Response-blind source norms.** Every criterion was written after seeing the responses. [proposition 145](#) shows contamination and redundancy are not separable without them.
2. **A second candidate set per person.** Required to test [theorem 111](#) out of sample and to resolve the probe-dependence question of §20(6).
3. **Typed fields.** 82.3% of criteria carry none ([table 4](#)); scope and exception loss are unstudiable here.
4. **Compilation lineage.** Absent, so §9’s three actions stay confounded.
5. *Y*. Not in the release.
6. **Cross-dataset / cross-domain.** One release exists.

### 22.2 Practical — could be run, was not

1. Held-out partitions for F1, F4, F5.
2. A specification curve over characteristic functions for the Shapley estimator; one was tested.
3. The bit curve of [section 9.1](#) across judges.
4. Two-way clustering ([theorem 85](#)) on the person-level quantities; only prompt-level clustering was applied, and annotators repeat across prompts.

### 22.3 Quantitative bounds that apply to every number here

| bound                                            | value                                 |
|--------------------------------------------------|---------------------------------------|
| judge attenuation ceiling                        | 0.741                                 |
| panel reliability ceiling                        | 0.9722                                |
| compounded ceiling on any correlation            | 0.720                                 |
| contamination-proxy variation in every flip rate | up to 1.8×                            |
| absolute winner claims: instrument share         | ≈ 40%                                 |
| absolute winner claims: presentation-order share | 8.0pp                                 |
| rubric corpus never joined                       | 1.8%                                  |
| design shortfall for 1pp resolution              | 968 of 1,213 prompts                  |
| pairwise-dependence screen: MDE at full $n$      | $ \phi  = 0.153$ against median 0.124 |

*Remark 152* (The last row read honestly). The median pairwise association in the operator grid is *below* the MDE at the largest available sample. The pairwise table is therefore a screen for *strong* dependence and is worded that way throughout; its nulls are underpowered rather than informative.

## 23 What a design that can measure this requires

Each row is derived from a specific result above, not proposed.

| requirement                                                        | derived from                                                | unlocks                                                                           |
|--------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------|
| response-blind source norms                                        | <a href="#">proposition 145</a>                             | the only route separating contamination from redundancy                           |
| $\geq 2$ candidate sets per person                                 | <a href="#">theorem 111</a> , <a href="#">corollary 101</a> | tests whether $M$ transfers; quantifies candidate-set overfitting                 |
| typed fields: veto, scope, exception, priority as separate columns | <a href="#">§8</a> , <a href="#">table 4</a>                | the index mismatch is a missing column, not a hard problem                        |
| interval-scale validation, or paired comparisons instead           | <a href="#">table 2</a>                                     | 11.7% of decisions currently rest on an unestablished assumption                  |
| $k > 4$ responses                                                  | <a href="#">theorem 94</a>                                  | raises the probe bound from 2 to $k - 2$                                          |
| $\geq 1,213$ prompts                                               | <a href="#">corollary 67</a>                                | 1pp resolution                                                                    |
| substantially more than 968 for pairwise structure                 | <a href="#">§22</a>                                         | the typical association is currently undetectable                                 |
| $\geq 3$ executors                                                 | <a href="#">§10.3</a>                                       | already available; the gauge is established                                       |
| compilation receipt (lineage)                                      | <a href="#">§9</a>                                          | separates selection, rewriting, weight-discarding                                 |
| declared $Q$ and declared $A$                                      | <a href="#">proposition 131</a> , <a href="#">remark 88</a> | without them “loss” is undefined and “defect” is inseparable from “social choice” |

## 24 Conclusion

The chain of [equation \(7\)](#) loses something at every arrow, and the losses are not of one kind.

- $\geq_i \rightarrow N_i$ . An unbounded relation is projected onto two free parameters ([theorem 94](#)) – forced, not recoverable. Within that, a person’s own criteria recover 39% of the direction of their own choice, and the individual component of that is 0.083 of cosine. The type structure is empty: 82.3% of criteria carry no condition, exception, priority, hedge or prohibition. And 11.7% of decisions depend on an interval reading of a scale that was never validated as one.
- $N_i \rightarrow G$ . Aggregation costs 0.1298 of normalised utility ([theorem 104](#)) – a social choice, but one made silently because  $A$  was never declared. The weighted-criteria route agrees with the Condorcet winner only 68.4% of the time while ranking rules agree 97–98%. 13.4% of pairwise relations flip when the option set changes. And 26.1% of the time the utilitarian winner is a response somebody called unacceptable.
- **Type**. No finite weight encodes a veto ([theorem 111](#)). This is the strongest statement in the paper and it required no instrument.
- $G \rightarrow C$ . Compilation changes 29.4% of decisions, with its three actions confounded. But the discarded magnitudes were worth about one bit: sign alone reproduces 89.8%.
- $C \rightarrow D$ . The executor’s reliability is 0.5497, capping every correlation at 0.741. Changing the reader changes 40% of winners; reordering the page changes 8%. Comparative claims survive all of it; absolute claims do not.
- $D \rightarrow Y$ . Does not exist in the release.

**Theory claim.** The losses inside the elicitation are larger than every loss the pipeline adds. The most defensible statement this object supports is a *theorem* about what a scalar scale cannot encode, and the most consequential one is that the next dataset should buy **fields and response-blindness**, not finer weights or more prompts.

## References